

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF INFORMATION COMMUNICATION TECHNOLOGY



SOICT

**Comprehensive Assessment of Air Quality
in Vietnam via Multi-Source Data Integration**

Instructor: Ph.D. Tran Nguyen Ngoc

Students: Trinh Hoang Anh - 20225470
Nguyen Tai Khoa - 20215066
Bui Khac Chien - 20220059

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Table of Contents

1	Introduction	4
1.1	Background and Motivation	4
1.2	Objectives	4
2	Data Sources and Acquisition	5
2.1	AQI and Meteorological Parameters Stations Data	5
2.1.1	The Official Monitoring Network (Envisoft) and OpenWeatherMap Meteorological Parameters	5
2.1.2	The Aggregator Network (IQAir)	6
2.1.3	Data Collection Strategy	6
2.2	Himawari-8 Aerosol Optical Depth (AOD)	6
2.2.1	Data Source and Specifications	7
2.2.2	Retrieval Caveats and Quality Information	8
2.2.3	Data Collecting Workflow	8
2.3	Additional Datasets	8
2.3.1	GIS Rainfall Dataset	9
2.3.2	MODIS Land Surface Temperature Dataset	9
2.3.3	Open Building and DEM Dataset	10
3	Data Preprocessing and Quality Control	10
3.1	Handling Missing Data	10
3.2	Time Series Validation and Anomaly Detection	11
3.3	Satellite Data Constraints and Spatiotemporal Sparsity	12
4	Data Analysis and Results	13
4.1	Cross-Source Consistency and Quality Control	13
4.1.1	Trend Alignment and Temporal Reliability	14
4.1.2	Algorithmic Divergence (The "Same Station" Phenomenon)	14
4.1.3	Identification of Data Artifacts ("Flat-Lining")	15
4.1.4	Strategic Dataset Selection	16
4.2	Spatio-Temporal Characteristics	16
4.2.1	Time Series Analysis	16
4.2.2	AQI Analysis by Hour of Day	17
4.2.3	Spatial Visualization	19
4.3	Multivariate Correlation Analysis	20
4.4	Correlation Analysis: Satellite AOD vs. Ground AQI	22
4.4.1	Level 2 (L2) Assessment: Broad Coverage with High Variance	22
4.4.2	Level 3 (L3) Assessment: High Precision in the North	23
5	Conclusion	24
5.1	Synthesis of Findings: The North-South Dichotomy	24
5.2	Evaluation of Monitoring Infrastructure	24
5.3	The Feasibility of Satellite Monitoring	24
5.4	Future Research Directions	25
A	AQI Multivariate Correlation	26

B	RANSAC and Final Linear Regression Fit	37
B.1	L2 RANSAC Analysis	37
B.2	L3 RANSAC Analysis	46

Abstract

This report presents a comprehensive assessment of air quality in Vietnam through the integration of multi-source ground station data with satellite remote sensing observations. We analyze the correlation between air quality indices (AQI/PM2.5) and meteorological parameters to understand pollution accumulation mechanisms, investigate temporal patterns including diurnal cycles, and compare pollution levels across different regions. The study cross-validates data from official monitoring networks (Envirosoft) and commercial aggregator platforms (IQAir), while also evaluating the relationship between ground-level measurements and Aerosol Optical Depth (AOD) data from the Himawari-8 satellite. Our findings reveal a fundamental North-South dichotomy in air quality dynamics, with Northern Vietnam exhibiting strong meteorological coupling through temperature inversion phenomena, while Southern regions demonstrate more stable atmospheric conditions. The satellite AOD analysis indicates high correlation with ground measurements during clear-sky conditions ($R^2 > 0.7$ for Level 3 products), suggesting potential for operational monitoring applications despite significant data sparsity challenges.

1. Introduction

1.1 Background and Motivation

In recent years, air quality has emerged as a critical public health concern in Vietnam, particularly in major urban centers and the Northern Delta region. Rapid urbanization and industrial activities have contributed to a significant rise in particulate matter concentrations, specifically PM2.5 and PM10. Recent environmental monitoring reports from early 2026 indicate that cities such as Hanoi and neighboring provinces frequently experience "Very Bad" to "Hazardous" air quality events, often ranking among the most polluted cities globally during peak episodes.

While anthropogenic factors, such as traffic emissions, construction, and industrial byproducts, are known contributors, the severity of pollution episodes is often exacerbated by meteorological conditions. Phenomena such as temperature inversion impede the dispersion of pollutants, creating a "trap" for particulate matter near the ground level. Consequently, identifying the precise drivers of pollution requires distinguishing between emission sources (e.g., rush-hour traffic) and meteorological parameters. Furthermore, the proliferation of diverse air quality monitoring sources necessitates a rigorous validation of data consistency to ensure accurate assessment.

Beyond ground-based observations, satellite-derived aerosol products offer a complementary perspective on air pollution dynamics. In this study, we also assess the relationship between Aerosol Optical Depth (AOD) retrieved from the Himawari-8 geostationary satellite and ground-level Air Quality Index (AQI) measurements across Vietnam. AOD provides column-integrated information on atmospheric aerosols and has the potential to enhance spatial coverage in regions with sparse monitoring stations. However, the extent to which satellite-based AOD reflects near-surface air quality under varying meteorological conditions remains uncertain. Evaluating the consistency and correlation between Himawari-8 AOD and AQI is therefore essential for understanding pollution mechanisms and for validating the applicability of satellite data in operational air quality assessment.

1.2 Objectives

This report aims to provide a comprehensive assessment of air quality by integrating multi-source ground station data with satellite remote sensing. The specific objectives of this study are:

- To evaluate meteorological dependency: Analyze the correlation between air quality indices (AQI/PM2.5) and meteorological parameters (such as humidity and temperature) to understand the impact of weather on pollution accumulation.
- To analyze temporal patterns: Investigate diurnal cycles of pollution to determine the correlation between pollution spikes and anthropogenic activities (e.g., traffic rush hours).
- To assess spatial distribution: Compare pollution levels across different areas within the same administrative boundaries to identify specific hotspots and regional trends.
- To cross-validate data sources: Assess the consistency and reliability of data by comparing colocated or proximal ground stations from different monitoring networks.
- To validate with remote sensing: Analyze the correlation between ground-level measurements and Aerosol Optical Depth (AOD) data from satellite platforms (Himawari-8) to evaluate the potential of satellite monitoring for wider coverage.

2. Data Sources and Acquisition

2.1 AQI and Meteorological Parameters Stations Data

To ensure a robust comparative analysis, this study utilizes two distinct AQI datasets representing different tiers of monitoring infrastructure. Both datasets cover a synchronized temporal window from mid-April to mid-December 2025, capturing the transition from the wet season into the early dry season. These datasets were cataloged into two separate station lists to facilitate cross-source validation.

2.1.1 The Official Monitoring Network (Envisoft) and OpenWeatherMap Meteorological Parameters

The primary dataset is derived from the Envisoft system, managed by the Vietnam Environment Administration (VEA).

Data Characteristics This source provides official data from government-operated fixed monitoring stations. As these stations undergo rigorous calibration, they serve as the reference standard for regulatory compliance and scientific validation in this study.

Spatial Scope Although the Envisoft network spans multiple provinces, this research prioritizes data from key administrative stations in major urban hubs, specifically Hanoi and Ho Chi Minh City, to analyze pollution dynamics in high-density environments.

Integration of Meteorological Parameters Because the Envisoft network focuses exclusively on air pollutants, it does not provide integrated meteorological parameters. To account for environmental catalysts such as wind speed, humidity, and temperature, atmospheric data was retrieved via the OpenWeatherMap API.

Data Merging Process Meteorological data points were selected based on the closest geographic coordinates to the Envisoft reference stations. These two streams (AQI and weather) were merged into a unified dataset, providing a comprehensive view of the interaction between pollutant concentrations and local climatic conditions.

2.1.2 The Aggregator Network (IQAir)

The second dataset is acquired from IQAir, a global air quality information platform.

Nature of Data IQAir serves as a mixed-source aggregator. It includes data from governmental stations but also integrates a wide network of low-cost commercial sensors (e.g., AirVisual Pro) deployed by organizations and individuals.

Spatial Scope A broader network of 15 selected stations across Vietnam, ensuring geographical diversity from Northern provinces (Hanoi, Thai Nguyen, Hai Duong) to Central (Hue, Quang Binh) and Southern regions (Vung Tau, Tra Vinh, Ho Chi Minh City).

Meteorological Parameters The IQAir dataset already contains meteorological data integrated within the platform.

2.1.3 Data Collection Strategy

Automated pipelines were developed to ingest data from both sources independently.

Collection Protocol

- **Envisoft Pipeline:** Dedicated scripts targeted the official portal to extract hourly readings from the selected administrative stations.
- **IQAir Pipeline:** A separate Python-based crawler interfaced with the IQAir API to retrieve real-time indices from the broader 15-station list.

Data Completeness and Synchronization Both pipelines were synchronized to run on an hourly schedule via GitHub Actions. To maximize utility, "Source ID" tags were maintained for every record, allowing the analysis team to distinguish between an official reading and a third-party sensor reading even when stations are geographically close (colocated).

Parameter Consistency Data was normalized to a common schema including local meteorological parameters.

2.2 Himawari-8 Aerosol Optical Depth (AOD)

To validate ground-based pollution trends over a broader spatial area and assess the potential of remote sensing for filling data gaps, this study utilizes both Level 2 (L2) and Level 3 (L3) Aerosol Property products from the Himawari-8/9 geostationary satellites, provided by the Japan Aerospace Exploration Agency (JAXA).

The aerosol retrieval algorithm is based on the theory of the scattering and absorbing of sunlight. The retrieval aims to minimize an object function and uses aerosol models. The retrieval method is derived from approaches developed by Higurashi and Nakajima (1998) and Fukuda et al. (2013). The JAXA Himawari products utilize retrieval algorithms based on upcoming Japanese Earth observation missions (GCOM-C, GOSAT-2, EarthCARE).

2.2.1 Data Source and Specifications

- **Source:** JAXA Himawari Monitor P-Tree System.
- **Product Versions:**
 - Level 2 (L2) ARP: High-frequency, high-resolution snapshots (10-minute intervals, 0.05° resolution) used to capture rapid diurnal variations.
 - Level 3 (L3) ARP: Spatiotemporally aggregated L2 data (hourly and daily averages, we only use hourly in this report) used to provide continuous coverage and mitigate noise inherent in instantaneous observations.
- **Temporal Range:** Data collection commences from January 2025.

Below are key aerosol properties provided by L2 and L3 ARP product:

Table 1: L2 ARP product parameters

Index	Description
AOT (500 nm)	Aerosol Optical Thickness
AE	Angstrom Exponent
AOT_uncertainty	Uncertainty of aerosol optical thickness
SSA	Single Scattering Albedo at 500 nm
RF	Optical depth ratio of fine mode
Location and Time	Latitude, longitude, and Hour (UTC)
QA_flag	Quality flag indicating various data conditions

Table 2: L3 ARP product parameters

Index	Description
AOT_Merged	Spatiotemporal interpolation of AOT_Pure
AOT_Pure	L2 AOT with strict cloud screening
AOT_L2_Mean	Average of L2 AOT for each pixel
AOT_L2_SDV	Standard deviation of AOT_L2_Mean within an hour
AOT_L2_Num	Total Number of L2 AOT within an hour
AOT_Merged_uncertainty	Uncertainty of AOT_Merged
AOT_Pure_uncertainty	Uncertainty of AOT_Pure
AE_Merged	Spatiotemporal interpolation of AE_Pure
AE_Pure	L2 AE with strict cloud screening
AE_L2_Mean	Average of L2 AE for each pixel
AE_L2_SDV	Standard deviation of AE_L2_Mean
AE_L2_Num	Total Number of L2 AE within an hour
QA_flag_merged	Quality flag of AOT/AE Merged
QA_flag_pure	Quality flag of AOT/AE Pure

2.2.2 Retrieval Caveats and Quality Information

Day/Night Limitation Aerosol retrieval is only possible during daytime because it relies on the scattering and absorbing of sun-light. Data is only retrieved when the solar zenith angle is less than 70 degrees. Consequently, there are no retrieved data in time transition periods between day and night.

Interpreting Uncertainty The uncertainty of AOT is reported using the parameter `AOT_uncertainty`. If using L3 products (which are derived from L2), the parameter `"AOT_Merged_uncertainty"` is unitless, and the uncertainty can be expressed as `"AOT_Merged" ± "AOT_Merged_uncertainty"`.

Quality Assurance Flag (QA_flag) The L2 ARP product includes a Quality Assurance Flag (`QA_flag`). This flag contains bit fields detailing various data conditions relevant to the retrieval quality.

2.2.3 Data Collecting Workflow

The data collection workflow is built around a custom Python-based pipeline designed for maximum data retention, departing from standard protocols that often discard "noisy" pixels during the initial download. By avoiding early-stage quality filtering, the pipeline preserves all retrieval records, ensuring that a comprehensive sensitivity analysis remains possible later in the study. The process begins with the ingestion of raw NetCDF files for both L2 and L3 products, which are then converted into GeoTIFF format. During this conversion, every pixel, regardless of uncertainty or quality flag status, is meticulously retained in the raw dataset.

To ensure computational efficiency without sacrificing detail, the global Full-Disk images undergo Geospatial Trimming. Using high-resolution shapefiles from the GADM database, the pipeline clips the imagery to the administrative boundaries of Vietnam. Following this, the system executes a Point-Based Colocation extraction by aligning coordinate metadata from established Envisoft and IQAir station datasets. The algorithm performs a raster pixel-wise extraction, capturing all relevant parameters specifically for the grid pixel corresponding to each ground station location for every available timestamp.

This "all-in" approach to data gathering culminates in a robust Stratification Strategy. Because no data is discarded during the extraction phase, the final dataset provides the flexibility to dynamically test various quality control thresholds during the modeling phase. Instead of being constrained by pre-filtered inputs, the analysis team can experiment with different parameters, such as comparing the results of filtering $AOT_{uncertainty} < 0.5$ versus < 1.0 , to determine the optimal balance between data volume and signal accuracy.

2.3 Additional Datasets

To provide a robust context for the primary analysis of AQI and AOD relationships, we have identified and curated several supplementary environmental datasets. While the current scope of this study prioritizes the Himawari-8 and ground-station analysis, the following datasets have been acquired to support future phases of research. These additional layers are critical for isolating meteorological catalysts (such as precipitation washout) and understanding the influence of urban morphology (topography and building density) on pollutant dispersion.

2.3.1 GIS Rainfall Dataset

The precipitation data is derived from the Integrated Multi-satellitE Retrievals for GPM (IMERG), which stands as NASA’s most sophisticated algorithm for generating global, high-resolution satellite-based precipitation estimates. For this study, we utilize the IMERG "Final" Run product, a research-quality dataset produced with a 3 to 4 month latency. As of early 2026, this ensures that the data (currently processed through late 2025) has been rigorously calibrated against ancillary sources, such as ground-based rain gauges. This calibration process provides a level of accuracy far superior to the "Early" or "Late" IMERG products, which are typically reserved for near real-time monitoring rather than formal scientific verification.

In terms of spatial and geographic characteristics, the dataset is provided at a 0.1° geographic resolution (approximately 10 km at the equator). The global grid consists of a 3600-column by 1800-row matrix, spanning from 90°S to 90°N and 180°W to 180°E . The spatial reference system used is EPSG 4326 (WGS84), employing a Plate Carree (Equidistant Cylindrical) projection. To ensure high temporal fidelity, the data is aggregated into 30-minute intervals, allowing for a granular analysis of precipitation accumulation and intensity over short durations.

For analysis, the stored integer values must be converted into physical units using a fixed scaling factor of 10. Specifically, for variables such as the precipitation rate (mm/h) or total accumulation (mm), the relationship is defined as:

$$\text{Physical Unit} = \frac{\text{Stored Integer}}{10}$$

For example, a raw pixel value of 100 represents a physical rate of 10.0 mm/h, while a value of 250 represents 25.0 mm of accumulation. The dataset distinguishes between liquid and ice phases, and it uses a specific value of 29999 to denote missing data. This comprehensive suite of variables, including liquid/ice rates and phase percentages, allows for a highly detailed characterization of wet deposition processes and atmospheric cleansing.

2.3.2 MODIS Land Surface Temperature Dataset

To assess surface thermodynamic conditions, we utilize the MODIS/Terra Land Surface Temperature/Emissivity Daily L3 Global 1km product (Shortname: MOD11A1). This Level-3 (L3) dataset is derived from the Moderate Resolution Imaging Spectroradiometer (MODIS) instrument aboard the Terra platform. The data is spatially manipulated into a gridded map projection using the Sinusoidal (SIN) Grid, where each tile comprises 1,200 rows by 1,200 columns. While the nominal resolution is 1 km, the actual grid size is precisely 0.928 km by 0.928 km. We utilize Collection 6.1 (C61), which maintains the same Hierarchical Data Format, Earth Observing System (HDF-EOS) structure as Collection 6 but incorporates significant enhancements to the Level-1B (L1B) input products and calibration changes to improve accuracy.

The daily per-pixel Land Surface Temperature (LST) values are constructed by mapping pixels from the Level-2 swath product (MOD11_L2) onto the Sinusoidal grid. The LST retrieval employs the generalized split-window algorithm, applied strictly under clear-sky conditions. In regions where multiple MODIS observations overlap, a frequent occurrence above 30 degrees latitude, the Collection 6.1 algorithm employs a specific selection logic to determine the representative pixel value. The algorithm prioritizes clear-sky LSTs obtained at smaller viewing zenith angles to minimize geometric distortion. However, if an observation at a larger zenith angle exhibits a temperature value higher by at least 2 Kelvin (K), it is selected instead, as higher temperatures generally indicate a lower likelihood of cloud contamination.

The MOD11A1 product contains several Scientific Data Sets (SDSs), primarily focusing on daytime and nighttime LST and Emissivity (bands 31 and 32), alongside essential geometry layers. Observation times (Day_view_time and Night_view_time) are recorded in Local Solar Time, calculated as UTC plus the longitudinal offset (degrees/15). Furthermore, the viewing geometry in Collection 6 products follows a specific sign convention: while the zenith angle represents the absolute angle from nadir, the Day_view_angle and Night_view_angle utilize a negative sign to indicate that the MODIS instrument is viewing the grid pixel from the east.

Data acquisition and archival are managed through the Level-1 and Atmosphere Archive and Distribution System Distributed Active Archive Center (LAADS DAAC). Files are identified using the standard Julian Date nomenclature (e.g. MOD11A1.AYYYYDDD.hXXvYY.061...), where YYYYYDDD represents the acquisition year and day, and hXXvYY denotes the specific horizontal and vertical tile identifiers. To focus the analysis on the study area, we restricted the download to the specific tiles covering Vietnam: h27v06, h28v06, h27v07, h28v07, and h28v08.

2.3.3 Open Building and DEM Dataset

To account for the physical "trapping" of pollutants within both urban and natural landscapes, the model integrates high-resolution morphological and topographic data. Urban density and street canyon effects are characterized using the Google Open Buildings dataset, which provides vector outlines derived from satellite imagery to estimate building height and concentration. By mapping these features, the model can assess how dense urbanization drives PM2.5 retention and traps local emissions at street level.

Complementing this, a Digital Elevation Model (DEM), sourced from the SRTM Global 1 arc second dataset, is used to analyze topographic relief at a 30-meter resolution. This allows for the identification of basin effects, which are particularly significant in Hanoi and the Northern Delta; in these areas, surrounding mountain ranges act as physical barriers that inhibit wind flow, causing pollutants to accumulate during stable meteorological conditions.

3. Data Preprocessing and Quality Control

Following data acquisition, a unified preprocessing pipeline was applied to ensure consistency across the Envisoft, IQAir, and Satellite datasets. Given the observational nature of the study, a minimal intervention strategy was adopted to preserve data integrity and avoid introducing synthetic bias through aggressive imputation.

3.1 Handling Missing Data

Although the automated crawler system minimized redundancy via timestamp-based duplicate checking, data gaps persisted due to occasional API connection failures. The following handling protocols were applied:

- **Prioritization of Pollutant Data:** AQI and PM2.5 records were retained even if auxiliary meteorological data (e.g., temperature, humidity) was missing, ensuring the primary variable of interest was not discarded.
- **Validity Filtering:** Records were strictly removed only if the primary AQI value or Timestamp was entirely absent or invalid.

- **Preservation of Gaps:** Missing quantitative values were encoded as NaN (Not a Number) rather than being imputed with means or medians. This allows for flexible handling in downstream analysis, such as ignoring nulls during correlation checks while preventing artificial smoothing in time-series plotting.

3.2 Time Series Validation and Anomaly Detection

To ensure statistical integrity for the spatiotemporal modeling, a specific filtering protocol was applied to the raw datasets:

- **Artifact Removal (Flat-lining):** To address "data logger freezing" artifacts observed in the aggregator network, sequences of identical values persisting for longer than 3 hours were flagged and excluded from the correlation and diurnal cycle datasets.
- **Retention for Quality Control:** Crucially, these artifacts were retained during the initial Quality Control assessment (Section 4.1) to accurately document and visualize the limitations of the monitoring infrastructure.
- **Range Consistency:** Data points exhibiting physically impossible values (e.g., negative concentrations) were globally removed.

3.3 Satellite Data Constraints and Spatiotemporal Sparsity

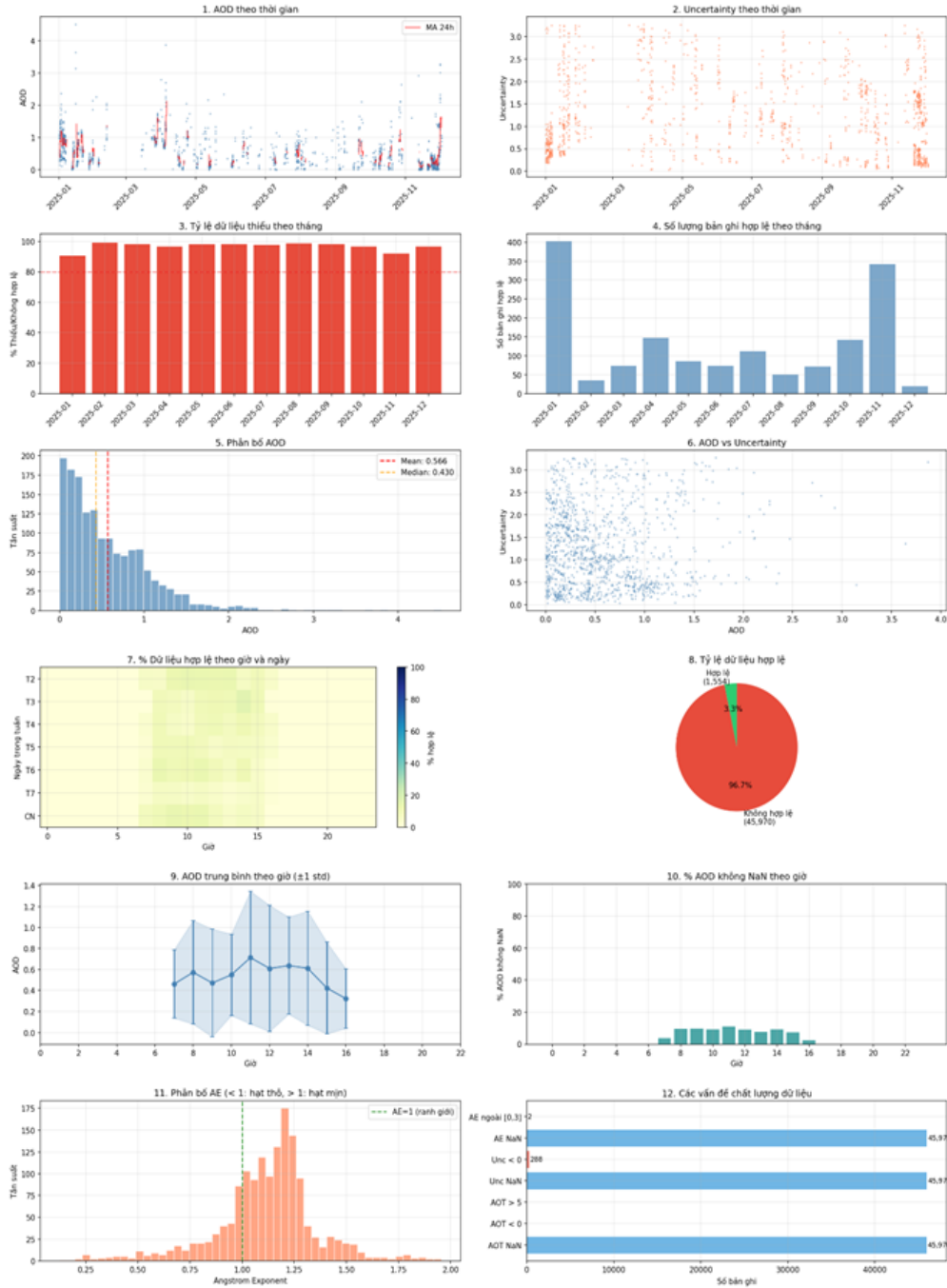


Figure 1: AOD Data Quality Report at HUST (Hanoi University of science and technology).

A significant limitation of this study is the extreme sparsity of the Aerosol Optical Depth (AOD) data retrieved from the Himawari-8 satellite.

- **Data Availability:** The valid data rate for the central Hanoi station (HUST) is only 3.3%, with 96.7% of the observation period lost to cloud cover or retrieval errors.

- **Regional Trends:** This challenge is consistent nationwide; no station in Vietnam exceeded a 5% valid data threshold. The highest availability was found in Southern regions (Long An at 4.9%), while the lowest occurred in Central and Northern provinces.
- **Analytical Scope:** Consequently, the AOD analysis must be interpreted strictly as a "clear-sky" assessment. The satellite is effectively "blind" during the heavily clouded or rainy periods that characterize the majority of the Vietnamese calendar year.

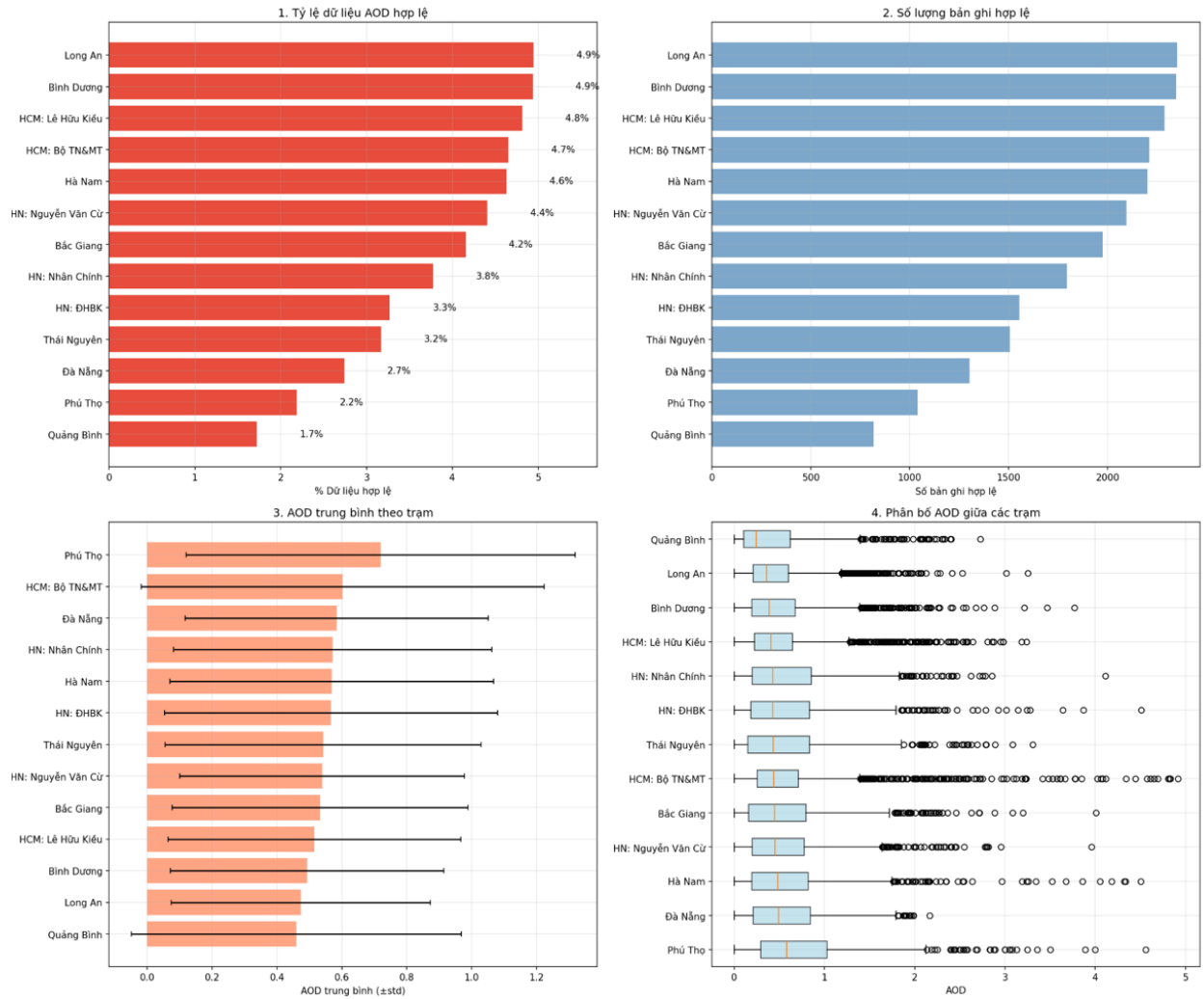


Figure 2: Comparison of AOD data quality across different monitoring stations, highlighting regional variations in data availability.

4. Data Analysis and Results

4.1 Cross-Source Consistency and Quality Control

Before integrating the datasets for spatiotemporal analysis, a rigorous validation assessment was performed to characterize the discrepancies between the official reference network (Envisoft) and

the commercial aggregator network (IQAir). This comparison focused on identifying systematic biases, algorithmic artifacts, and data reliability issues.

4.1.1 Trend Alignment and Temporal Reliability

A pairwise comparison of the Le Huu Kieu administrative station (Envisoft) and the Saigon Pearl commercial sensor (IQAir), located approximately 3 km apart, reveals a robust temporal alignment.

Both datasets exhibit identical inflection points during major pollution events (e.g., the severe episodes in November and December). This confirms that the aggregator platform preserves the fundamental temporal structure of the pollution signal, validating its utility for analyzing regional diurnal and seasonal trends.

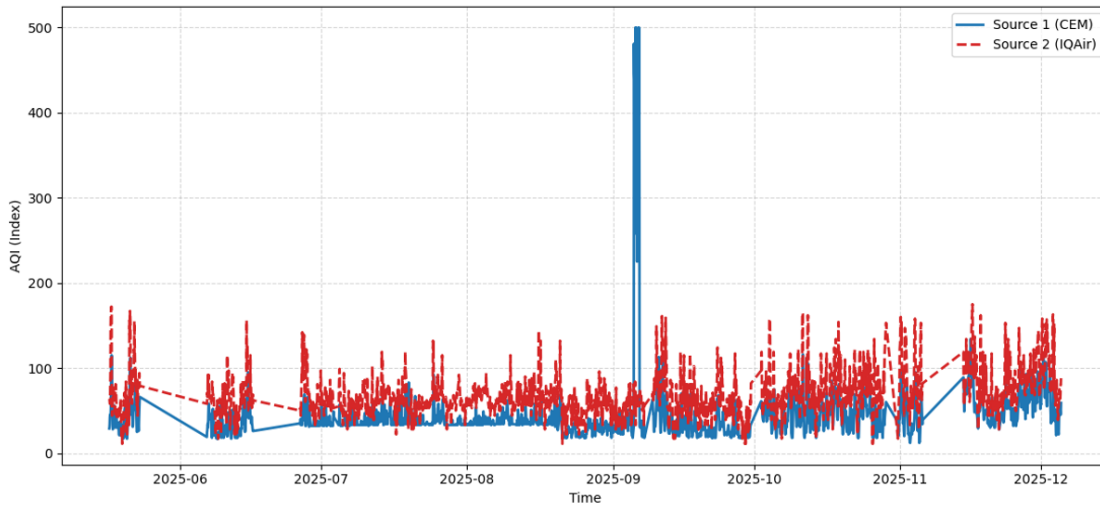


Figure 3: Comparative time-series of Air Quality Index (AQI) between the Le Huu Kieu administrative station (Source 1: CEM) and the Saigon Pearl commercial sensor (Source 2: IQAir) in Ho Chi Minh City.

4.1.2 Algorithmic Divergence (The "Same Station" Phenomenon)

A critical discrepancy was observed where reported values differed significantly even when both platforms cited the identical data source. Specifically, at the HUST Parabol station, the aggregator (IQAir) explicitly attributes the "Northern Center for Environmental Monitoring" (Envisoft) as its source, yet reports values that systematically diverge from the official government portal.

Observed Phenomenon:

- **Mass Concentration Bias (PM2.5):** The aggregator consistently reports lower raw mass concentrations than the official portal ($\text{IQAir} < \text{Envisoft}$). This suggests the application of a negative correction factor to the raw feed.
- **Index Bias (AQI):** Conversely, the aggregator consistently reports higher Air Quality Index values ($\text{IQAir} > \text{Envisoft}$).

Attribution of Bias: This divergence is attributed to post-processing algorithms employed by the aggregator rather than sensor error.

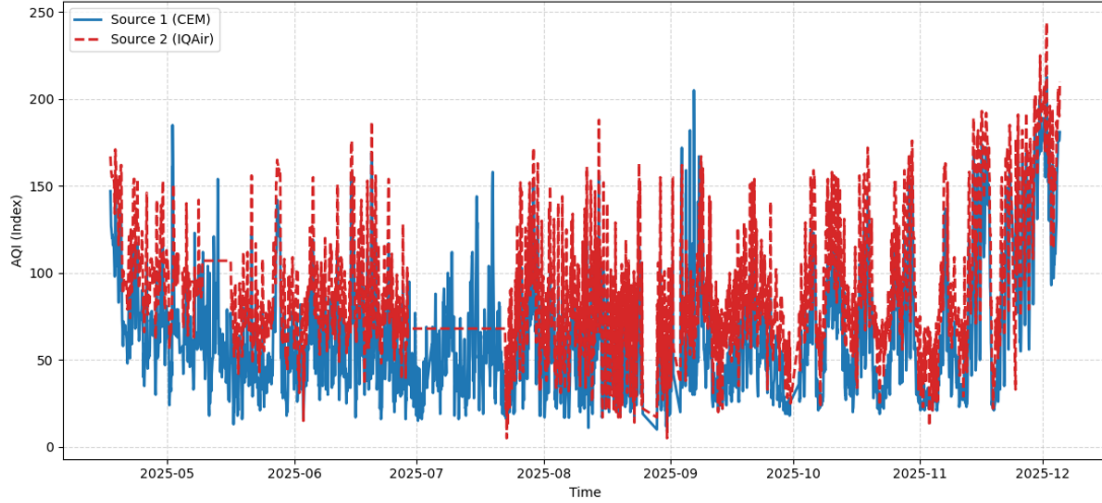


Figure 4: Comparative time-series of Air Quality Index (AQI). The aggregator data (Red) reports systematically higher values than the official source (Blue) despite using the same input data. This offset is caused by the aggregator's use of the stricter US EPA calculation standard, contrasting with the QCVN 1459:2019 standard used by the government portal.

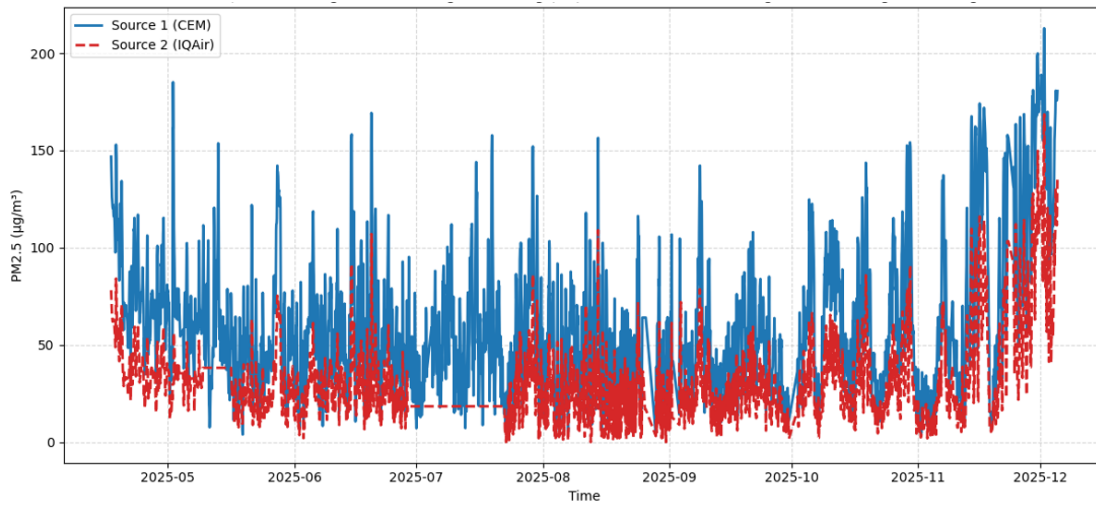


Figure 5: Comparison of PM2.5 concentrations. In contrast to the AQI, the aggregator's raw mass concentrations (Red) are consistently lower than the official reference (Blue). This negative offset suggests the application of an algorithmic humidity correction factor by the aggregator to subtract water vapor mass from the raw government reading.

4.1.3 Identification of Data Artifacts ("Flat-Lining")

A distinct limitation identified in the aggregator dataset is the prevalence of "Abnormal Recording" periods. The aggregator data frequently exhibits extended intervals (e.g., in late July and mid-August) where the recorded value remains mathematically constant for days.

- **Implication:** These "flat-line" artifacts typically indicate a data logger freezing event or a loss of connection where the last known value is repeated indefinitely.

- **Mitigation:** To ensure statistical integrity, a pre-processing filter was applied to the IQAir dataset for the subsequent analysis. Sequences of identical values persisting for longer than 3 hours were flagged as artifacts and excluded from the spatiotemporal modeling.

4.1.4 Strategic Dataset Selection

Given the variations in post-processing algorithms and the occasional recording gaps observed in the aggregator network, the remainder of this study (from Section 4.2 onwards) utilizes the official Envisoft dataset as the primary reference. While the aggregator network proved highly effective for validating broad temporal trends, the Envisoft network provides the standardized consistency necessary for the detailed meteorological and satellite-based comparisons presented in the subsequent sections.

4.2 Spatio-Temporal Characteristics

4.2.1 Time Series Analysis

Northern Region (Hanoi, Bac Ninh, Bac Giang) Stations in the North exhibit a clear seasonal shift. From April to September, the Air Quality Index (AQI) remains relatively low, fluctuating mostly between 50 and 100. Starting in late October and continuing through December, there is a sharp and sustained increase in AQI levels. During this period, values frequently exceed 150 (Unhealthy), forming a distinct upward trend at the end of the year.

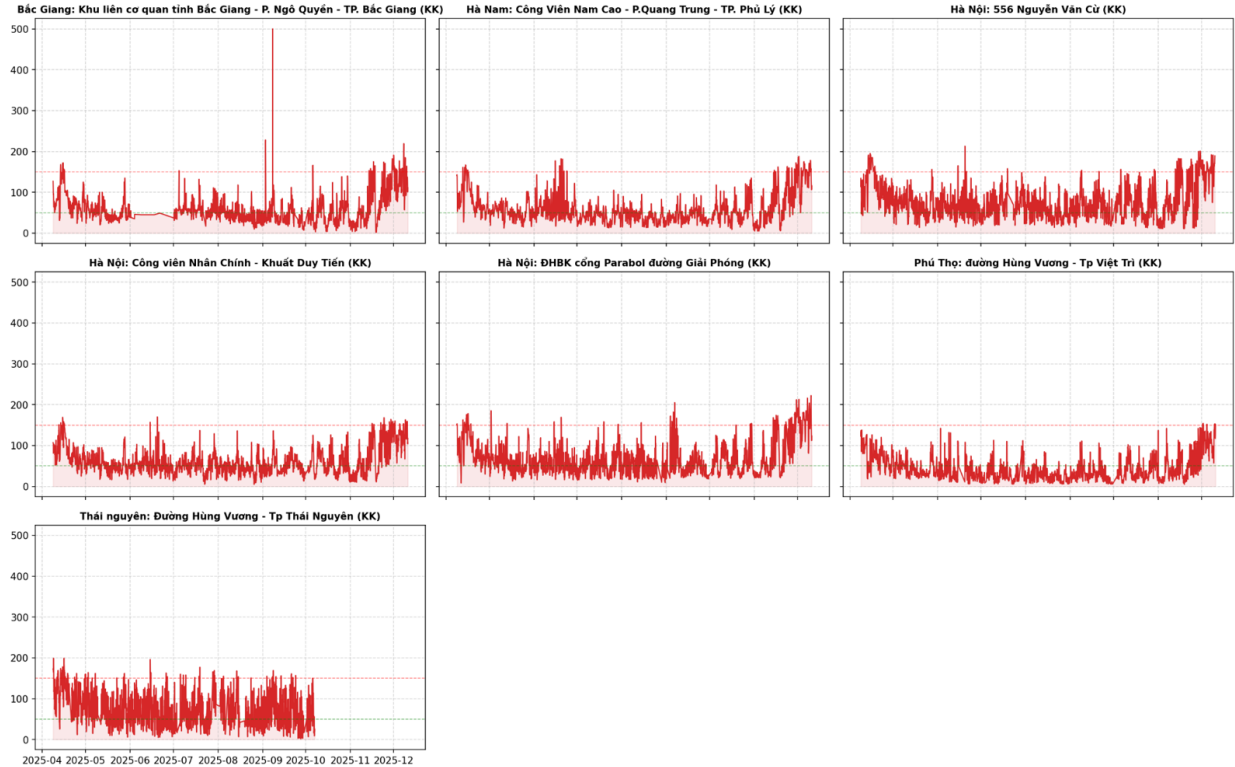


Figure 6: Regional Air Quality Analysis – North Vietnam (AQI Time Series).

Southern Region (Ho Chi Minh City, Binh Duong) In contrast, the Southern stations show a more consistent pattern. The AQI levels remain relatively stable throughout the observation

period. While there are occasional daily fluctuations, there is no massive jump in pollution levels during November and December compared to the mid-year months.

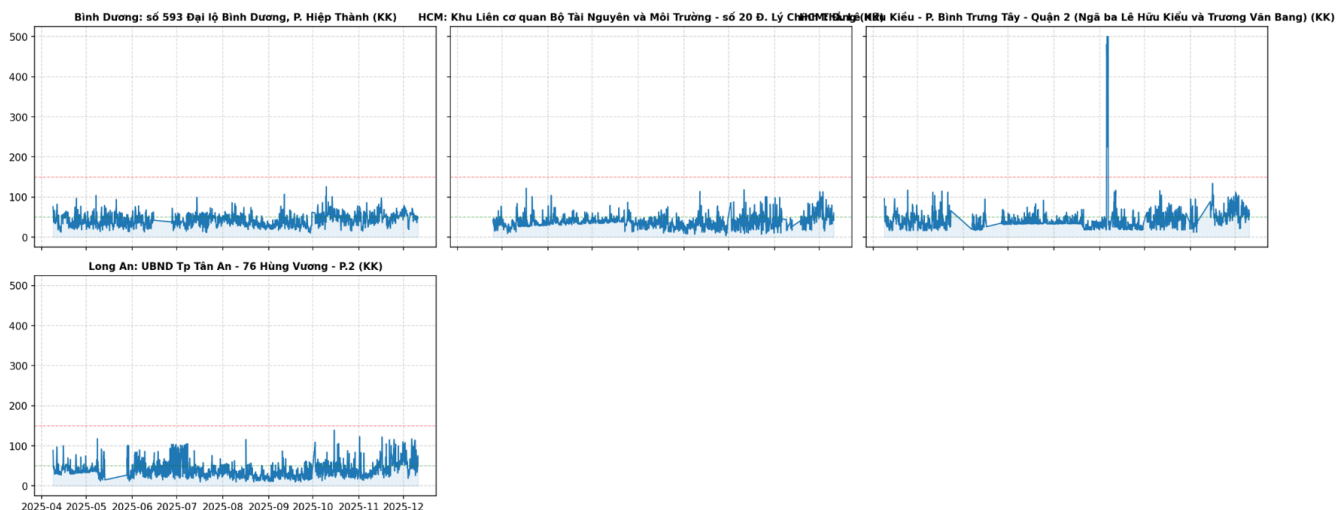


Figure 7: Regional Air Quality Analysis – South Vietnam (AQI Time Series).

Central Region (Da Nang, Quang Binh) The Central region shows a mixed pattern. While there are some individual spikes in pollution, the overall trend does not show the prolonged high-pollution "blocks" seen in the North during the end of the year.

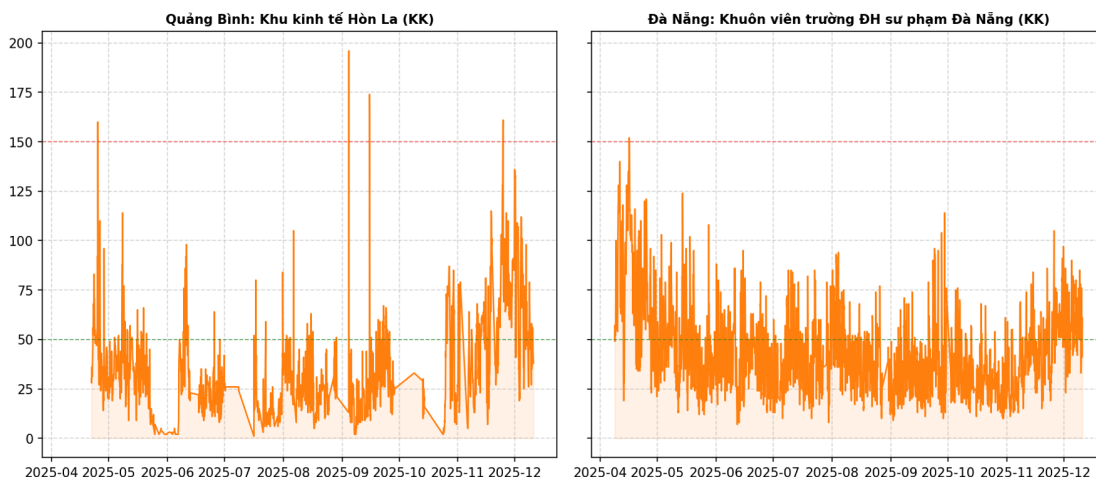


Figure 8: Regional Air Quality Analysis – Central Vietnam (AQI Time Series).

4.2.2 AQI Analysis by Hour of Day

Northern Region A comparison of hourly trends reveals a distinct shift in daily pollution patterns between the seasons. By contrasting the diurnal profiles of July and December, two different behaviors are observed.

The Summer Pattern: Morning Peak (July) During the summer months, the daily profile follows a clear "morning pulse." A distinct peak in AQI is observed centering around 08:00 AM.

This timing aligns closely with the morning commute, suggesting that in the summer, air quality is primarily influenced by fresh emissions from rush-hour traffic.

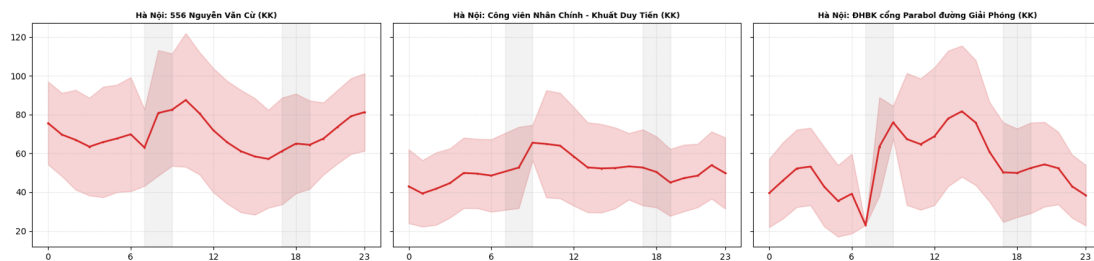


Figure 9: Diurnal AQI pattern in North Vietnam during summer (July), showing morning rush-hour peak.

The Winter Pattern: Nighttime Accumulation (December) In sharp contrast, the winter profile shows a completely different cycle. The highest pollution levels are not found during the day, but rather late at night, specifically building up between 22:00 and 05:00. Since traffic volume is low during these hours, this accumulation suggests that stable weather conditions at night are likely trapping pollutants near the ground, overriding the usual traffic signals seen in the summer.

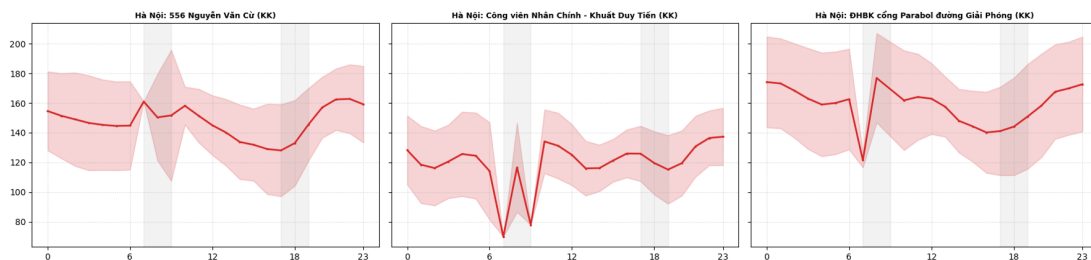


Figure 10: Diurnal AQI pattern in North Vietnam during winter (December), showing nighttime accumulation.

Southern Region Unlike the Northern region, which shifts to a dangerous nighttime pollution cycle in winter, the Southern region (Ho Chi Minh City) shows a completely different behavior. Comparing the daily profiles of July and December reveals a much more stable environment.

The Summer Pattern: Stable and Flat (July) In the summer month of July, the pollution profile is remarkably flat. There is no discernable peak at any specific time of day. The line remains relatively level from morning to night. This suggests that during the wet season, the air quality in Ho Chi Minh City is consistent and does not experience sharp spikes from traffic or weather changes during the day.

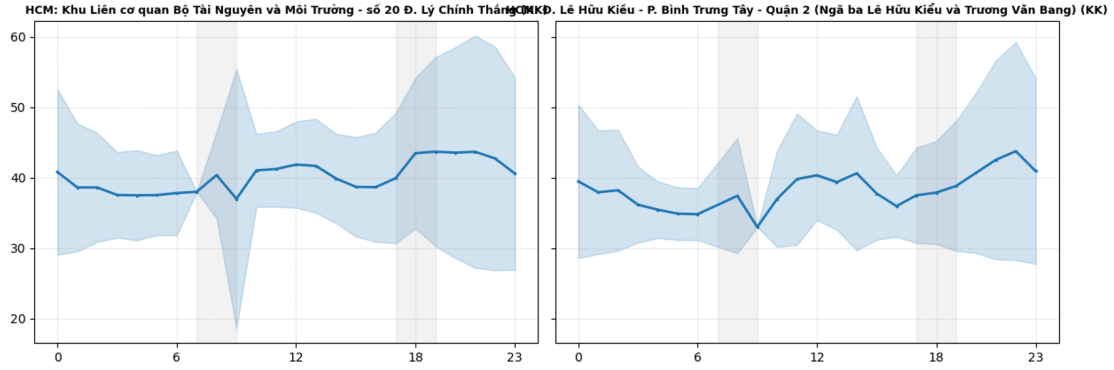


Figure 11: Diurnal AQI pattern in South Vietnam during summer (July), showing stable profile.

The Winter Pattern: Midday Rise (December) In December, the pattern changes slightly, but it is the opposite of Hanoi. Instead of peaking at night, the pollution levels tend to rise during the day, reaching their highest point around noon (10:00 to 14:00). Crucially, the air quality at night does not rise like the north.

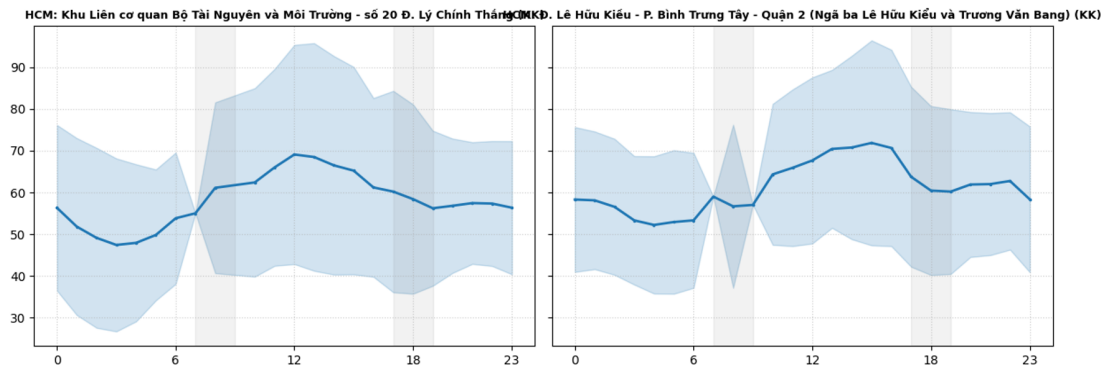


Figure 12: Diurnal AQI pattern in South Vietnam during winter (December), showing midday peak.

4.2.3 Spatial Visualization



Figure 13: Correlation Matrix at.

4.3 Multivariate Correlation Analysis

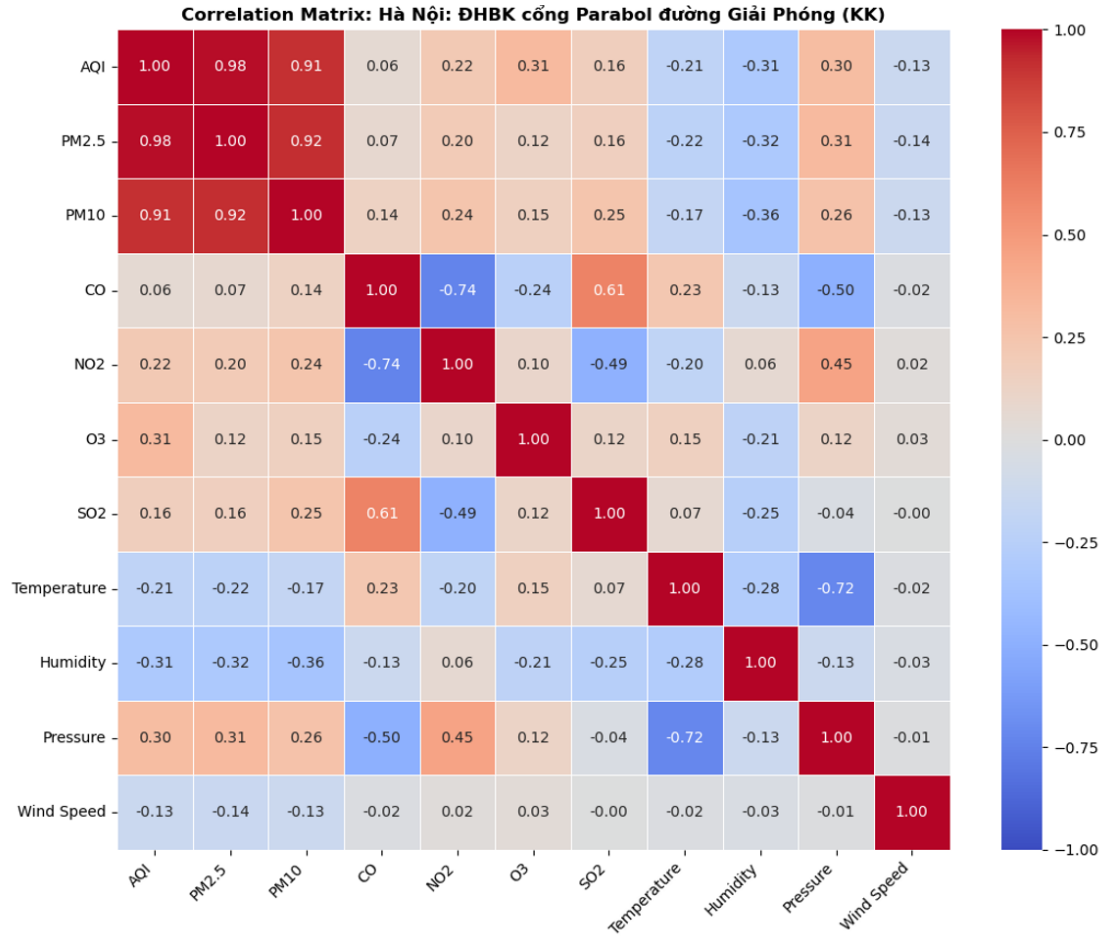


Figure 14: Correlation Matrix for a Representative Northern Monitoring Station (Hanoi University of Science and Technology).

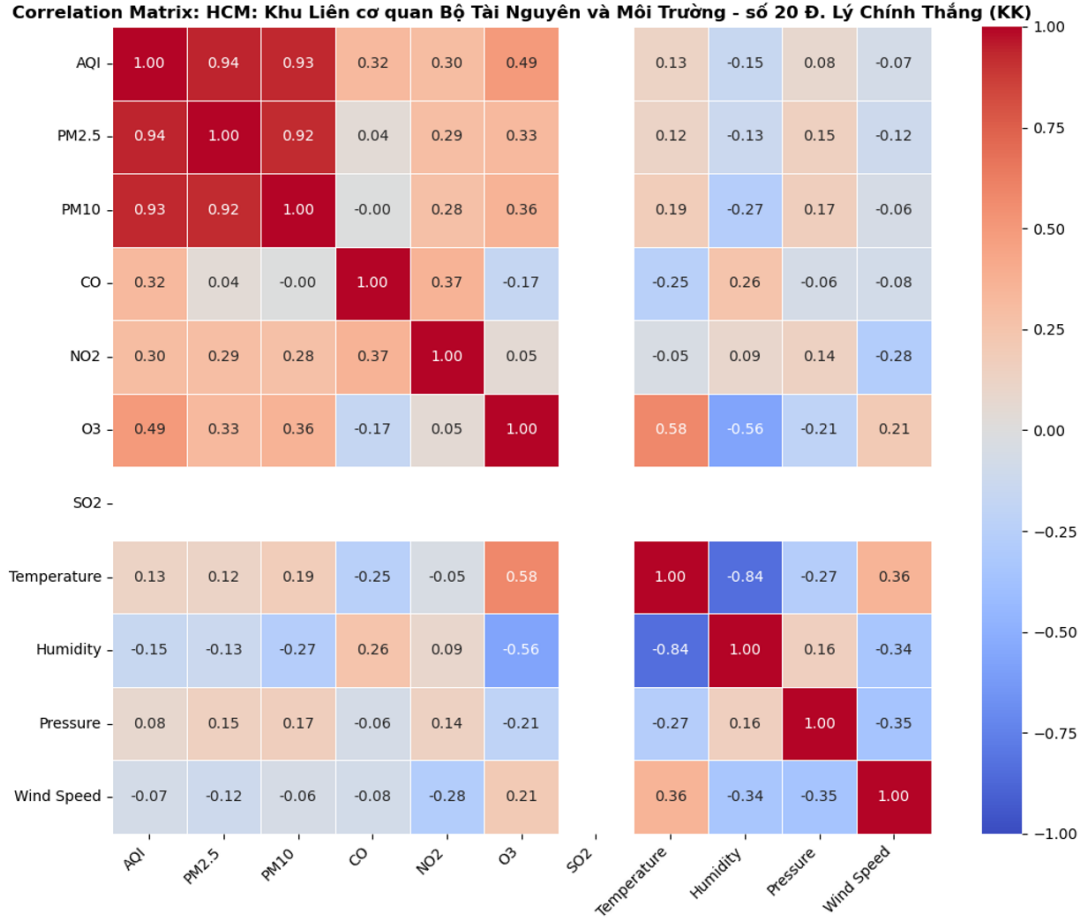


Figure 15: “Correlation Matrix Between AQI, Air Pollutants, and Meteorological Variables at a Representative Southern Station (Ho Chi Minh City – Ministry of Natural Resources and Environment, 20 D. Ly Chinh Thang).

A comparative analysis of the correlation matrices reveals a fundamental divergence: air quality in the Northern region appears noticeably more sensitive to meteorological fluctuations than in the South.

The Pressure "Trap" (Northern Specific) In Hanoi, Atmospheric Pressure exhibits a noticeable positive correlation with AQI ($r \approx +0.28$). This suggests that high-pressure systems play a distinct role in pollution accumulation for the capital, likely by stabilizing the atmosphere and reducing vertical mixing. In contrast, Ho Chi Minh City shows a negligible correlation ($r \approx +0.09$), indicating that the Southern basin is not subject to this same "barometric trapping" mechanism.

Temperature Response The regions show opposing sensitivities to temperature. Hanoi displays a discernible negative correlation ($r \approx -0.20$), consistent with the observation that colder days (Winter) are prone to higher pollution. Conversely, Ho Chi Minh City displays a weak positive correlation ($r \approx +0.12$), implying that cold-air stagnation is not a relevant driver in its tropical climate.

Humidity While humidity acts as a cleaning agent in both regions, the effect is more pronounced in the North (e.g., $r \approx -0.32$ at HUST Parabol). This reflects the sharp contrast between the North's dry, polluted winters and its wet, clean summers, a seasonal distinction that is less extreme in the Southern data.

Conclusion: The Role of Temperature Inversion Collectively, these coefficients indicate that the Northern atmosphere is "meteorologically constrained." The specific combination of high pressure ($r \approx +0.28$) and negative temperature correlation ($r \approx -0.20$) in Hanoi serves as a statistical signature for Temperature Inversion. These inversions create a stable atmospheric cap that suppresses vertical mixing, preventing pollutant dispersion.

This conclusion aligns directly with previous empirical research by Trinh et al. (2018) [1], who documented that temperature inversion is a primary driver of particulate matter accumulation in Hanoi, particularly during the winter months. In contrast, the Southern region remains relatively decoupled from these specific traps, driven instead by consistent tropical convection that prevents the formation of prolonged stable layers.

4.4 Correlation Analysis: Satellite AOD vs. Ground AQI

Following the quality assessment, a correlation analysis was performed to evaluate the predictive capability of Himawari-8 Aerosol Optical Depth (AOD) against ground-measured Air Quality Index (AQI). The analysis compared the raw Level 2 (L2) product against the aggregated Level 3 (L3) product.

Crucially, this analysis excluded ground data where $AQI < 50$. This filter was applied to remove the "clean air baseline," where low aerosol concentrations typically result in low signal-to-noise ratios for the satellite sensor, potentially skewing the regression with non-physical noise.

4.4.1 Level 2 (L2) Assessment: Broad Coverage with High Variance

The Level 2 dataset offers the widest geographic coverage but exhibits significant variability in performance due to the lack of temporal aggregation.

Regional Performance Performance varied widely by location. The station at Thai Nguyen (Hung Vuong Street) showed the most consistent L2 results, achieving an R^2 of 0.70 (ALL threshold) with a robust sample size ($N=50$). This indicates that L2 data can effectively track pollution in provincial areas. Conversely, stations in Hanoi and Ho Chi Minh City initially showed weak correlations in the "ALL" category ($R^2 < 0.1$) due to urban noise.

Threshold Selection (ALL vs. U05) The impact of uncertainty filtering was inconsistent across stations.

- **Effective Cases:** For Hanoi (556 Nguyen Van Cu), applying the stricter U05 filter drastically improved the correlation from negligible ($R^2 = 0.048$) to significant ($R^2 = 0.586$).
- **Ineffective Cases:** For Ho Chi Minh City (Le Huu Kieu St.), the U05 filter degraded performance (R^2 dropped to 0.099), whereas the unfiltered "ALL" data maintained a solid R^2 of 0.636.
- **Judgment:** Unlike L3 data, there is no single "best" filter for L2; "ALL" generally preserves the most usable data, while "U05" is necessary for specific noisy urban stations.

Table 3: L2 AOD vs. Ground AQI Correlation Results

Region	City	Filter	R^2	N	Inlier%
North	Hanoi	U05	0.59	70	63.6
North	Hanoi	U05	0.26	62	64.6
North	Thai Nguyen	ALL	0.70	50	56.8
North	Ha Nam	U15	0.25	119	55.1
Central	Quang Binh*	U05	0.97	9	42.9
South	Ho Chi Minh	ALL	0.64	50	53.2
South	Long An	U05	0.64	19	61.3

*Statistically insignificant due to small sample size

Table 4: L3 AOD vs. Ground AQI Correlation Results

Reg.	Station	R^2	N	In.%
N	Hanoi (Nguyen Van Cu)	0.719	35	77.8
N	Hanoi (Hanoi University of Science and Technology)	0.708	35	79.5
N	Hanoi (Nguyen Chinh)	0.618	43	75.4
N	Ha Nam	0.875	18	64.3
N	Bac Giang	0.727	30	58.8

Sample Size Sensitivity Caution is required when interpreting high correlations. For instance, Quang Binh (Economic Zone) produced a near-perfect R^2 of 0.974 at the U05 threshold. However, this model relies on only 9 inliers, rendering it statistically insignificant. Generally, the RANSAC algorithm retained 50 to 65% of L2 data as valid, reflecting the inherent noise in raw retrievals.

4.4.2 Level 3 (L3) Assessment: High Precision in the North

The Level 3 product, available primarily for the Northern region due to cloud constraints, demonstrated superior stability and accuracy.

Enhanced Reliability L3 data consistently outperformed L2 without requiring complex filtering. The "ALL" threshold yielded excellent results, rendering stricter filters largely redundant.

- Ha Nam (Cong Vien): Achieved the highest reliability with an R^2 of 0.875.
- Hanoi (556 Nguyen Van Cu): Unlike the L2 data, the L3 product showed strong performance even without strict filtering ($R^2 = 0.719$).
- Bac Giang (Lien Co Area): Maintained a robust correlation ($R^2 = 0.727$).

Data Purity The L3 dataset exhibited much lower noise levels. RANSAC inlier retention rates were consistently high, reaching 79.5% at Hanoi (HUST). This confirms that the aggregated L3 product successfully filters out the majority of the transient noise affecting the L2 dataset.

Summary: Level 2 data is useful for broad coverage but requires careful, station-specific filtering to handle noise. Level 3 data, while spatially limited, provides a highly accurate and "ready-to-use" proxy for air quality in the Northern region ($R^2 > 0.7$).

5. Conclusion

5.1 Synthesis of Findings: The North-South Dichotomy

This study has established a fundamental divergence in the air quality dynamics between Northern and Southern Vietnam, driven primarily by meteorological constraints rather than solely by emission intensity.

The Northern Trap The analysis confirms that air quality in the Northern Delta (Hanoi, Bac Giang, Ha Nam) is strictly coupled with meteorological conditions. The strong correlation between high pressure, low temperature, and pollution accumulation ($r \approx -0.20$ for temperature) validates the prevalence of Surface Temperature Inversions (STI). These inversions act as a "lid" during winter months, trapping pollutants regardless of traffic fluctuations, resulting in the distinct nocturnal pollution cycles observed in December. This finding aligns with previous empirical research by Trinh et al. (2018) [1], who documented that temperature inversion is a primary driver of particulate matter accumulation in Hanoi during winter.

The Southern Stability Conversely, Ho Chi Minh City and the Southern region exhibit a decoupled atmospheric state. With negligible correlations to pressure ($r \approx +0.09$) and a consistent tropical convective profile, the South avoids the "barometric traps" of the North. Pollution here is driven more directly by source emissions (traffic/industry) rather than seasonal meteorological stagnation.

5.2 Evaluation of Monitoring Infrastructure

The cross-validation between official (Envisoft) and aggregator (IQAir) networks revealed critical insights for operational monitoring. The foundational data utilized for these comparative analyses was derived from the Envisoft analysis report by Pham (2025) [2] and the IQAir analysis report by Hoang (2025) [3].

Temporal Reliability The aggregator network successfully captures the timing of pollution events (onset and duration), making it a viable tool for public trend awareness.

Systematic Bias However, significant algorithmic biases were identified. The aggregator consistently reports lower mass concentrations (PM_{2.5}) but higher Index values (AQI) compared to the official reference, likely due to different humidity correction factors and Index calculation standards (US EPA vs. QCVN).

Recommendation Low-cost sensor networks and aggregators should be used for temporal trend analysis but require localized calibration factors before being used for quantitative exposure assessment or regulatory compliance.

5.3 The Feasibility of Satellite Monitoring

The assessment of Himawari-8 AOD products highlights a "Quality vs. Quantity" trade-off:

The Cloud Limitation The primary barrier to operational satellite monitoring in Vietnam is extreme data sparsity. With valid data availability below 5% for most stations, passive satellite sensors are effectively "blind" during the critical winter monsoon and rainy seasons.

High-Precision Proxy However, during clear-sky conditions, the Level 3 (L3) AOD product proves to be an exceptionally accurate proxy for ground-level pollution in the North ($R^2 > 0.7$).

Operational Role Himawari-8 should not replace ground stations but rather serve as a spatial extrapolator during clear days, allowing authorities to map the extent of regional haze episodes extending beyond the limited ground station network.

5.4 Future Research Directions

To address the limitations identified in this study, future research should focus on:

1. **Gap-Filling Algorithms:** Developing Machine Learning models (e.g., Random Forest or LSTM) to impute missing satellite data by integrating the meteorological parameters (humidity, wind speed) and IMERG rainfall data identified in Section 2.3.
2. **Algorithm Harmonization:** Establishing a standardized correction protocol for low-cost sensors (IQAir) to align their raw mass concentration readings with the Envisoft reference standard.
3. **Targeted Policy Interventions:** Given the "Nighttime Accumulation" pattern in Northern winters, traffic restrictions or emission controls implemented during the day may be less effective. Policy efforts should focus on reducing background emissions (industry, construction) that accumulate under the nocturnal inversion layer.

References

- [1] T. T. Trinh, T. T. Trinh, T. T. Le, T. D. H. Nguyen, and B. M. Tu, "Temperature inversion and air pollution relationship, and its effects on human health in Hanoi City, Vietnam," *Environmental Science and Pollution Research*, vol. 25, no. 34, pp. 34698–34707, 2018.
- [2] N. H. N. Pham, "Analysis of Air Pollution Levels and the Impact of Meteorological Factors in Major Vietnamese Cities Based on Online Data," Project Report, School of Information and Communications Technology, Hanoi University of Science and Technology, 2025.
- [3] V. T. Hoang, "Air Quality Analysis," Project Report, School of Information and Communications Technology, Hanoi University of Science and Technology, 2025.
- [4] A. Higurashi and T. Nakajima, "Development of a Two-Channel Aerosol Retrieval Algorithm on a Global Scale Using NOAA AVHRR," *Journal of the Atmospheric Sciences*, vol. 55, no. 6, pp. 924–941, 1998.
- [5] S. Fukuda et al., "New approaches to removing cloud shadows and evaluating the 380 nm surface reflectance for improved aerosol optical thickness retrievals from the GOSAT/TANSO-Cloud and Aerosol Imager," *Journal of Geophysical Research: Atmospheres*, vol. 118, no. 24, pp. 13520–13531, 2013.

A. AQI Multivariate Correlation

This appendix presents the correlation matrices of Air Quality Index (AQI) measurements across all monitoring stations. The matrices illustrate the pairwise relationships between stations, highlighting the degree of similarity and spatial coherence in AQI variations. These results provide supporting evidence for regional pollution patterns and help assess the consistency and interdependence of AQI observations among different locations.

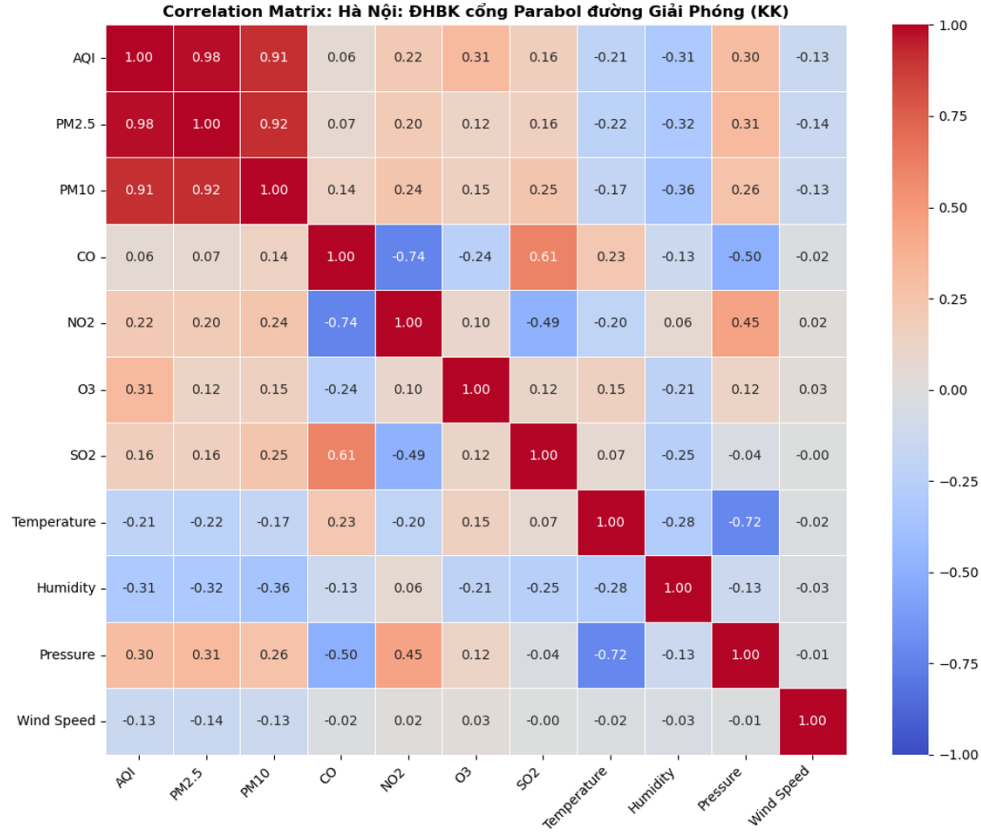


Figure 16: Correlation matrix for Ha Noi (ĐHBK) Station.

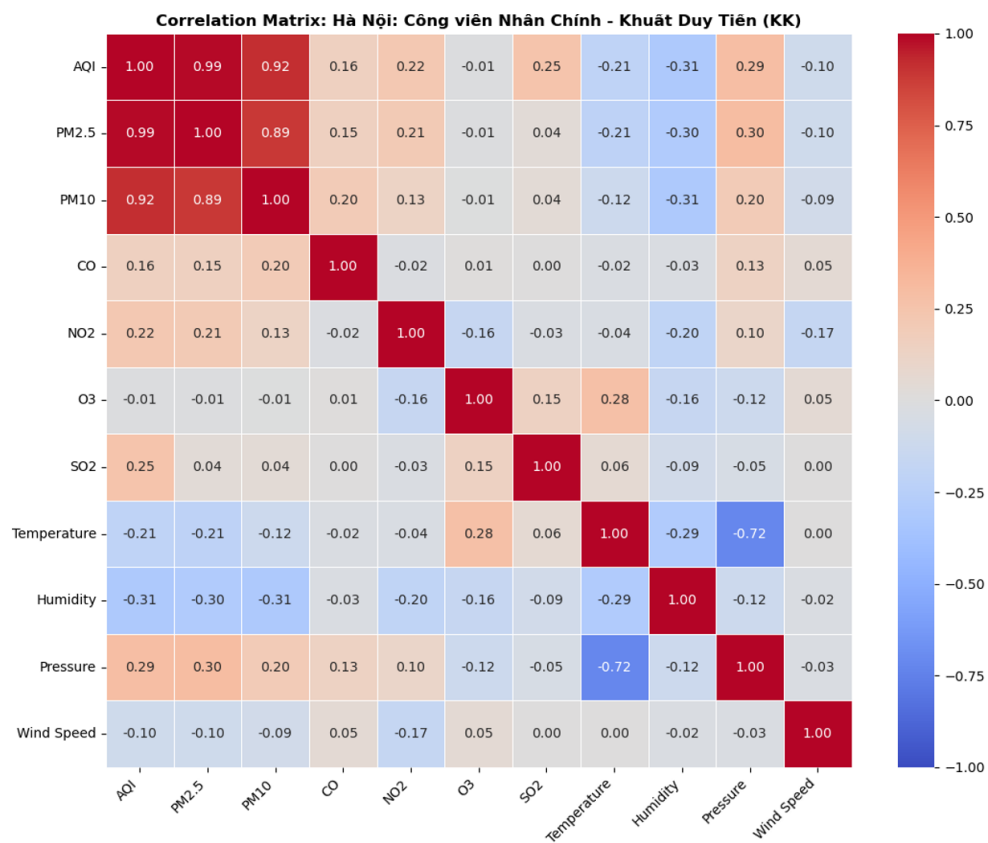


Figure 17: Correlation matrix for Ha Noi (Nhan Chinh park) Station.

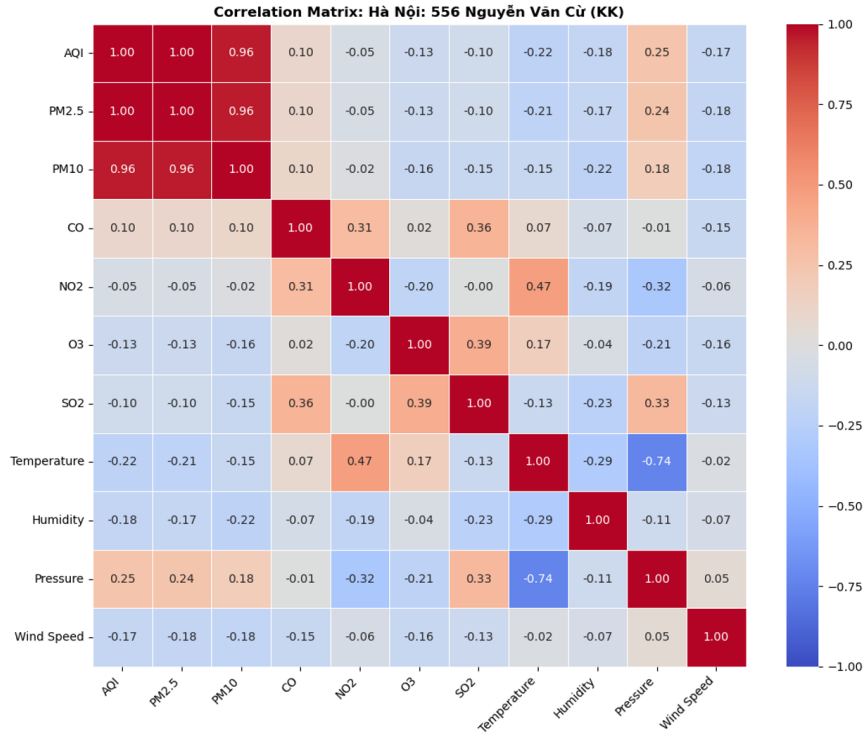


Figure 18: Correlation matrix for Ha Noi (Nguyen Van Cu) Station.

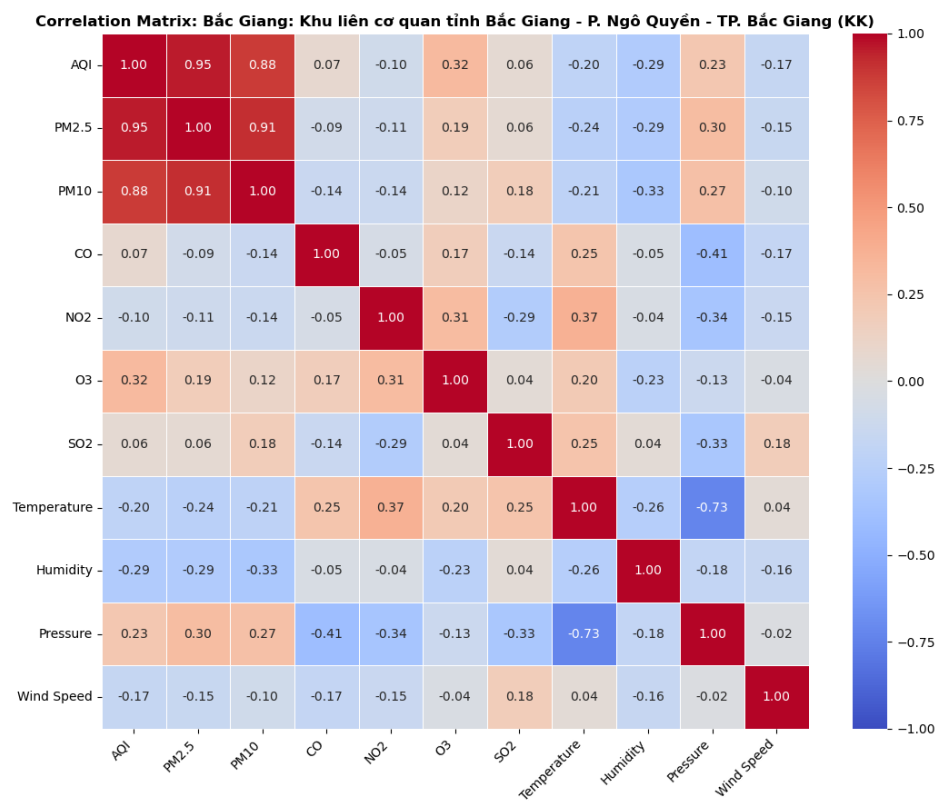


Figure 19: Correlation matrix for Bac Giang Station.

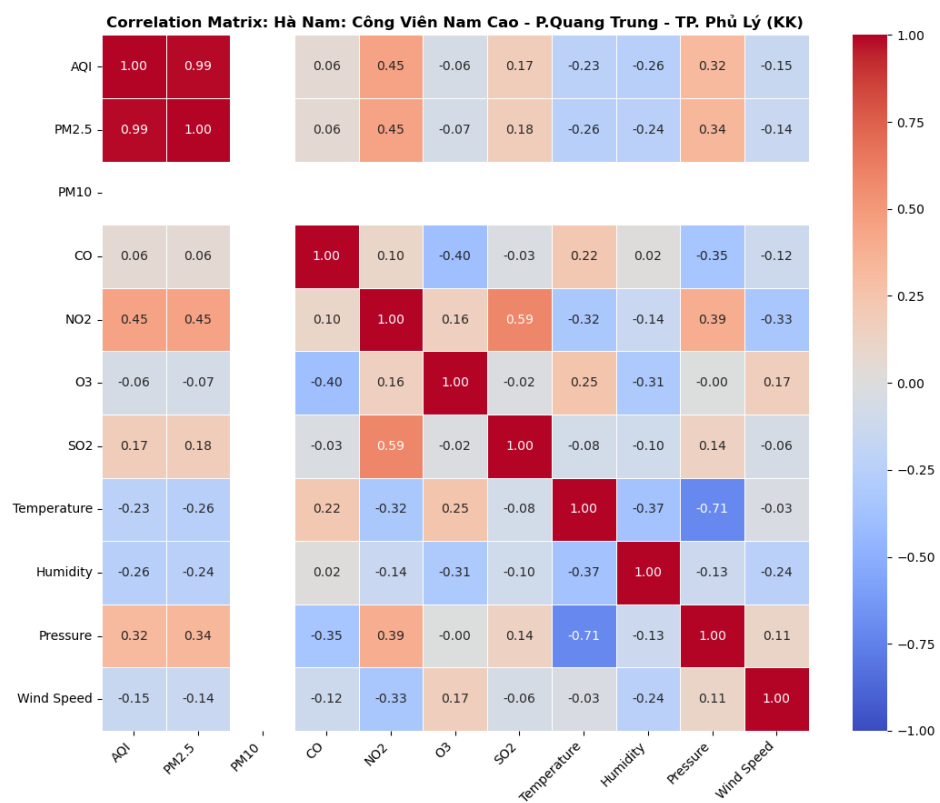


Figure 20: Correlation matrix for Ha Nam Station.

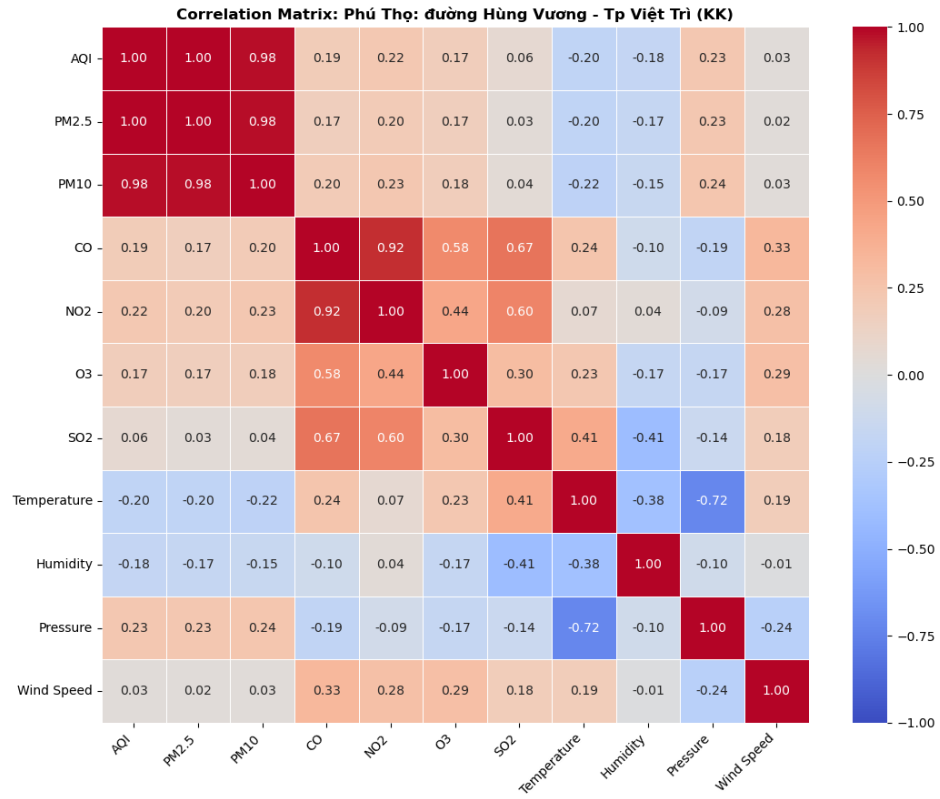


Figure 21: Correlation matrix for Phu Tho Station.

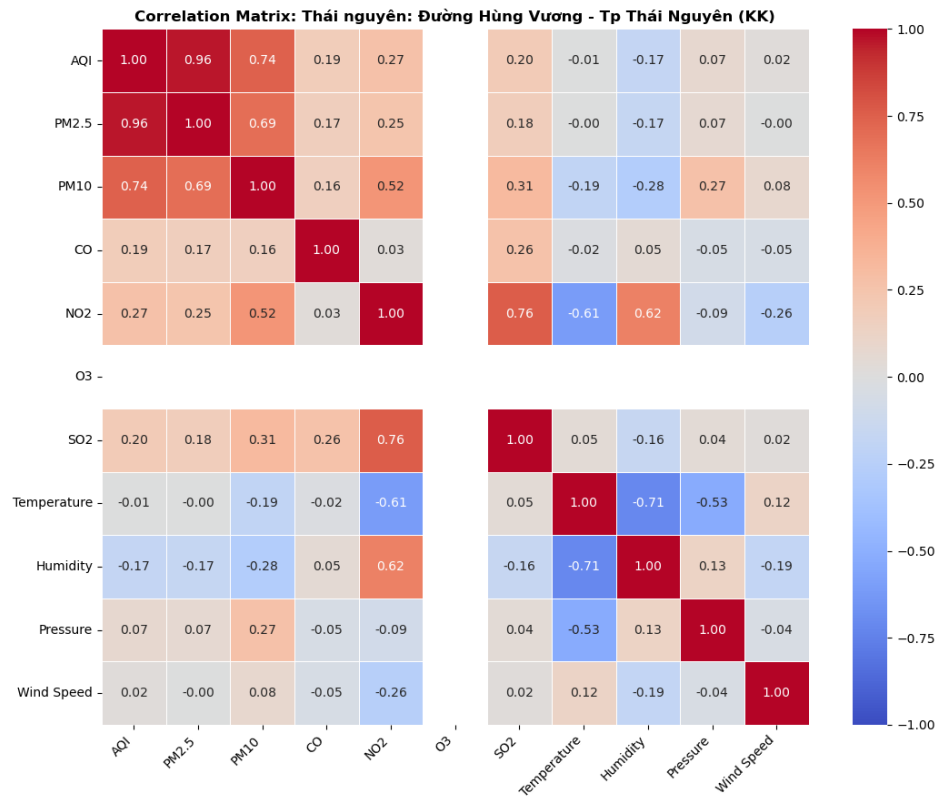


Figure 22: Correlation matrix for Thai Nguyen Station.

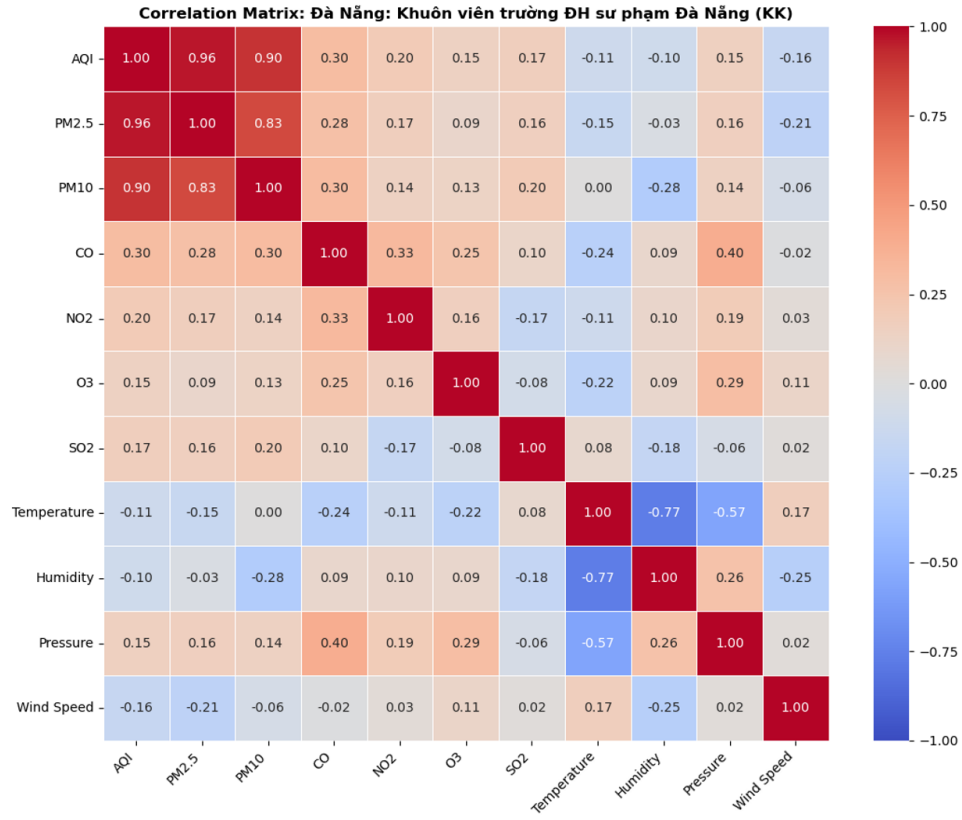


Figure 23: Correlation matrix for Da Nang Station.

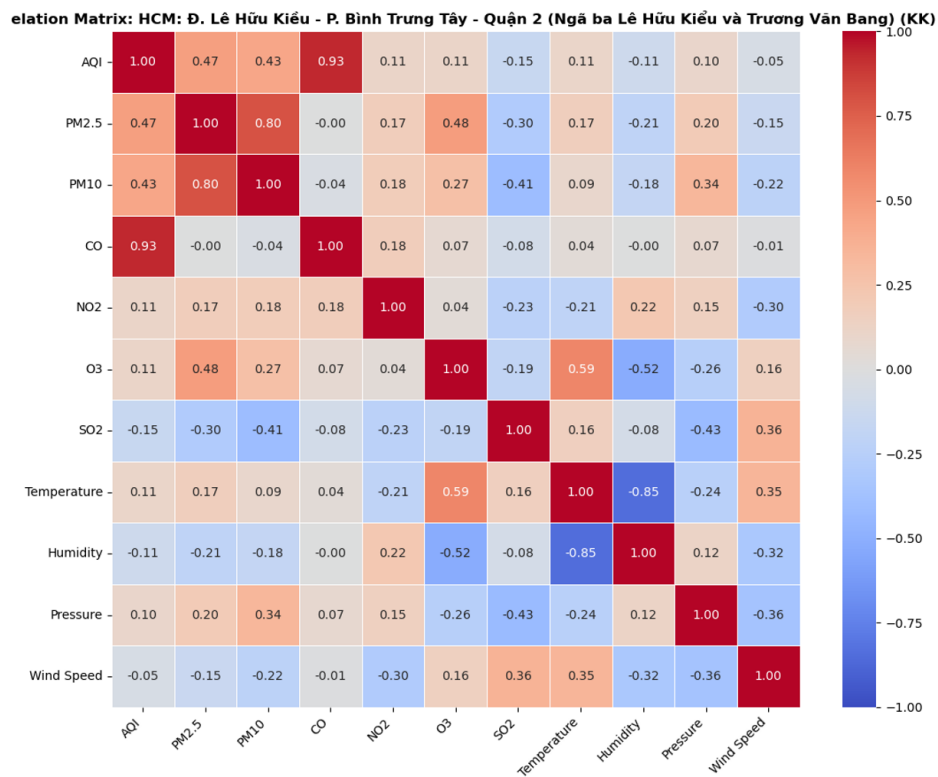


Figure 24: Correlation matrix for HCM (Le Huu Kiem) Station.

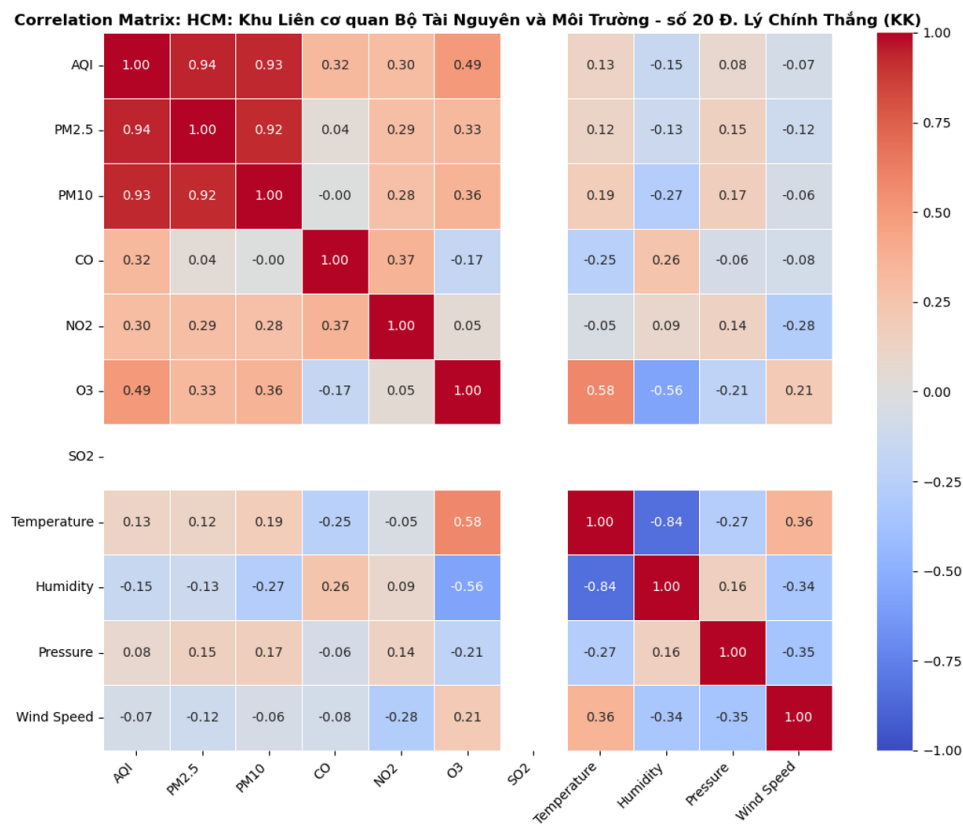


Figure 25: Correlation matrix for HCM (Khu liên cơ quan Bộ Tài Nguyên và Môi Trường - số 20 Đ. Lý Chính Thắng (KK)) Station.

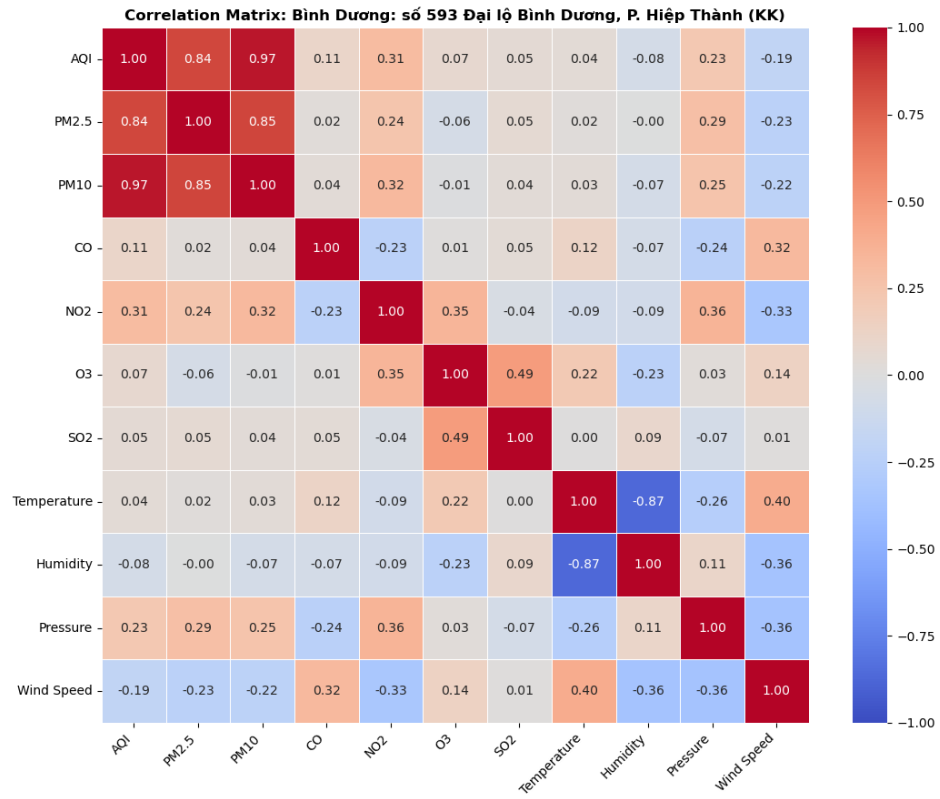


Figure 26: Correlation matrix for Binh Duong Station.

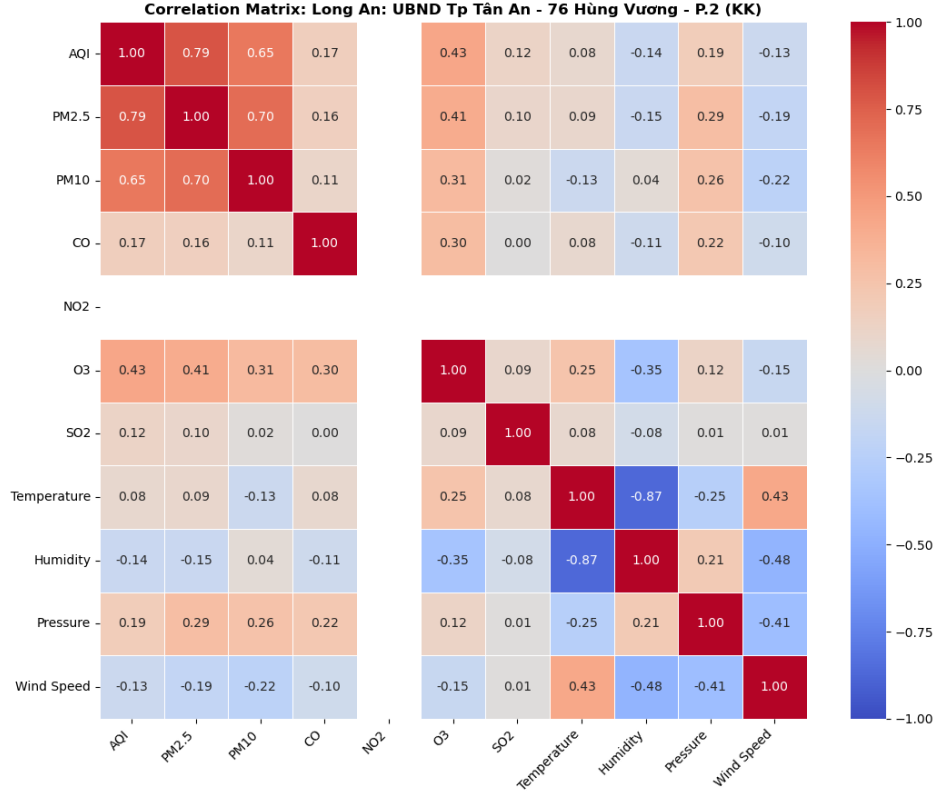


Figure 27: Correlation matrix for Long An Station.

B. RANSAC and Final Linear Regression Fit

We applied the RANSAC algorithm to detect and exclude outliers, then fitted an ordinary least squares (OLS) linear regression to the RANSAC-identified inliers. The analysis was performed on both L2 and L3 datasets across multiple stations.

B.1 L2 RANSAC Analysis

The Level 2 dataset required robust outlier detection due to the high noise levels inherent in instantaneous satellite retrievals. The RANSAC algorithm was configured with a residual threshold optimized for each station's noise characteristics. Inlier retention rates varied from 42.9% to 64.6%, reflecting the challenging nature of L2 data quality.

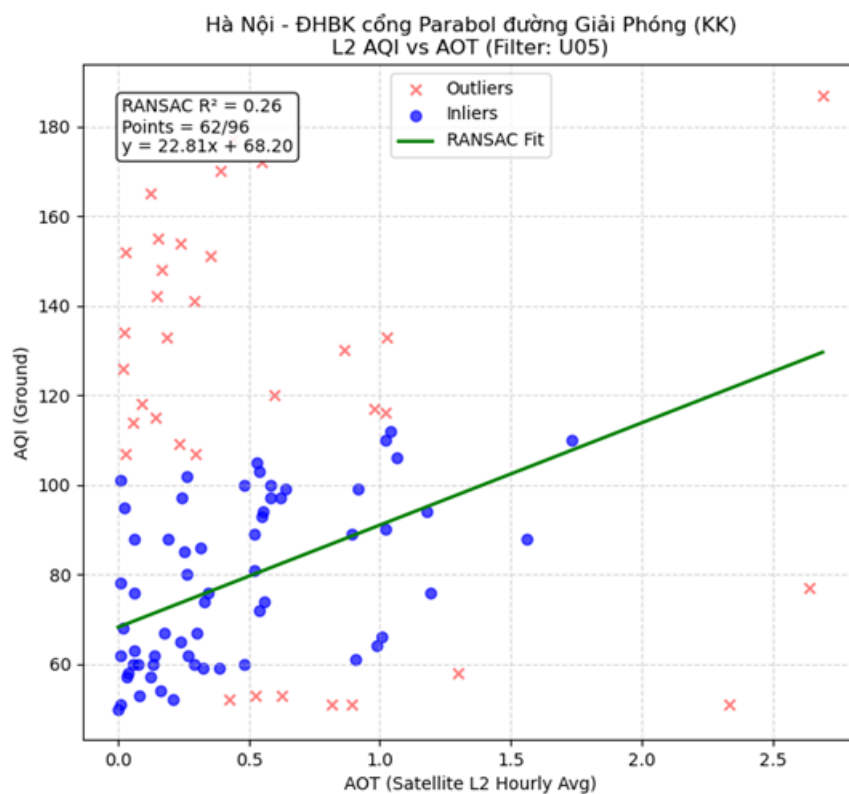


Figure 28: L2 RANSAC regression fit for Station 1.

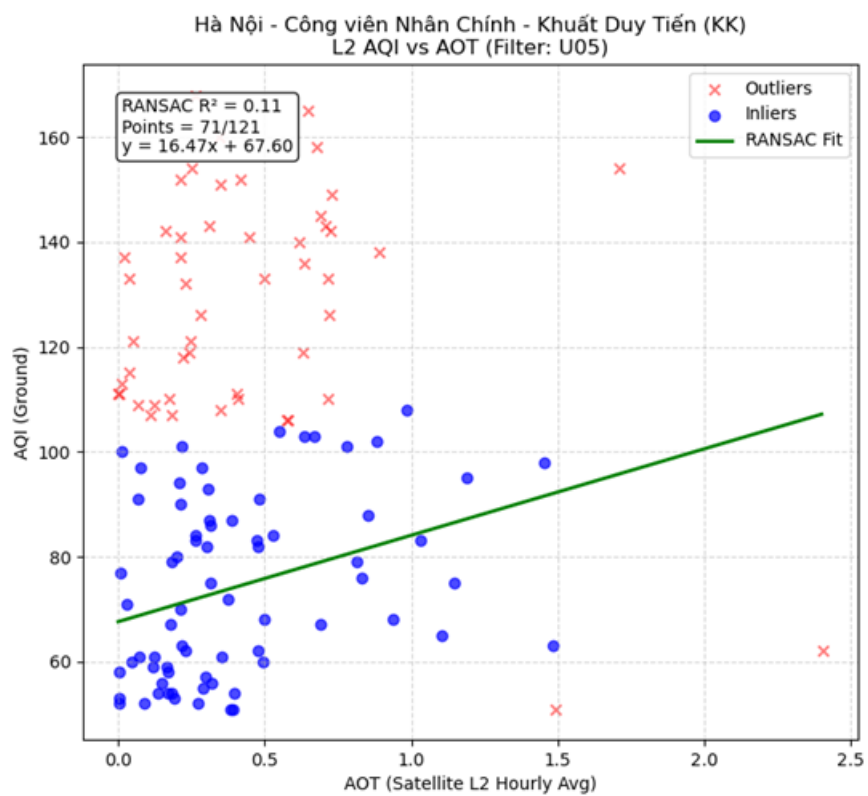


Figure 29: L2 RANSAC regression fit for Station 2.

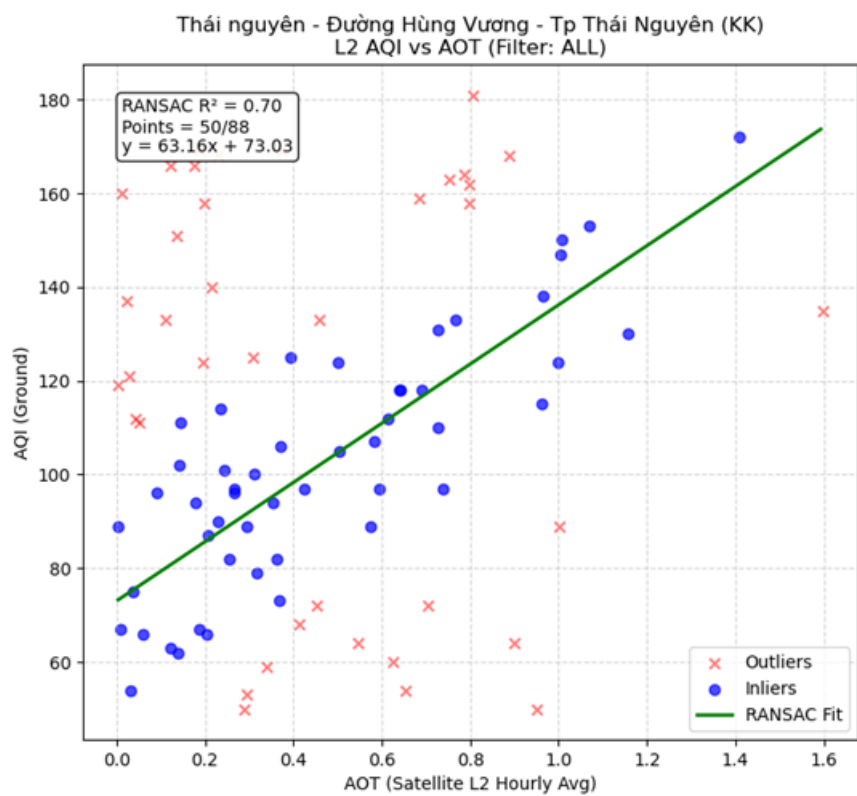
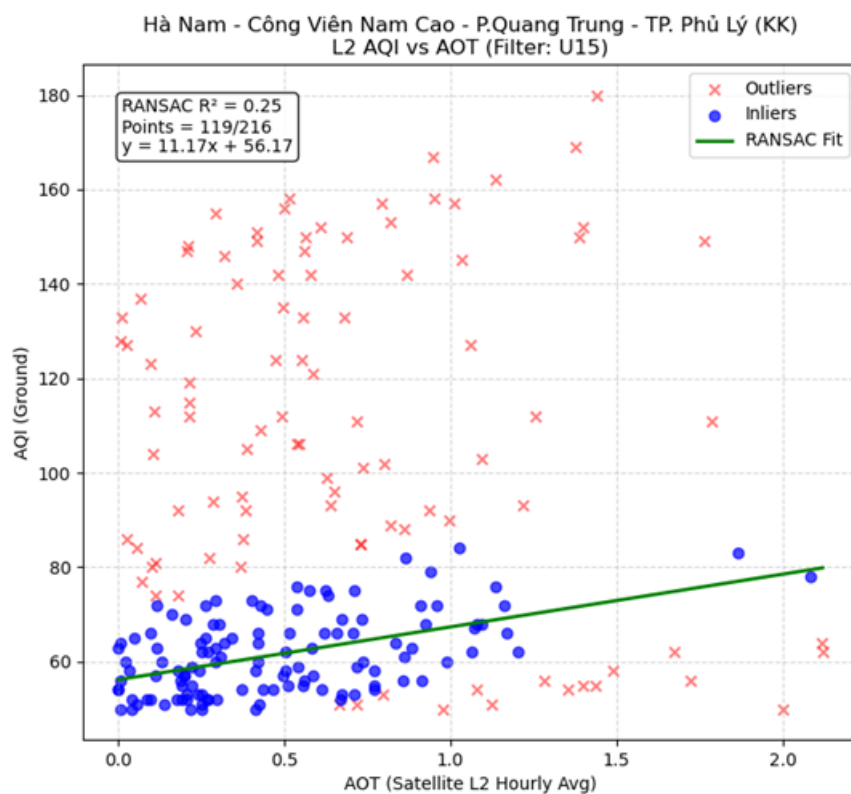


Figure 30: L2 RANSAC regression fit for Station 3.



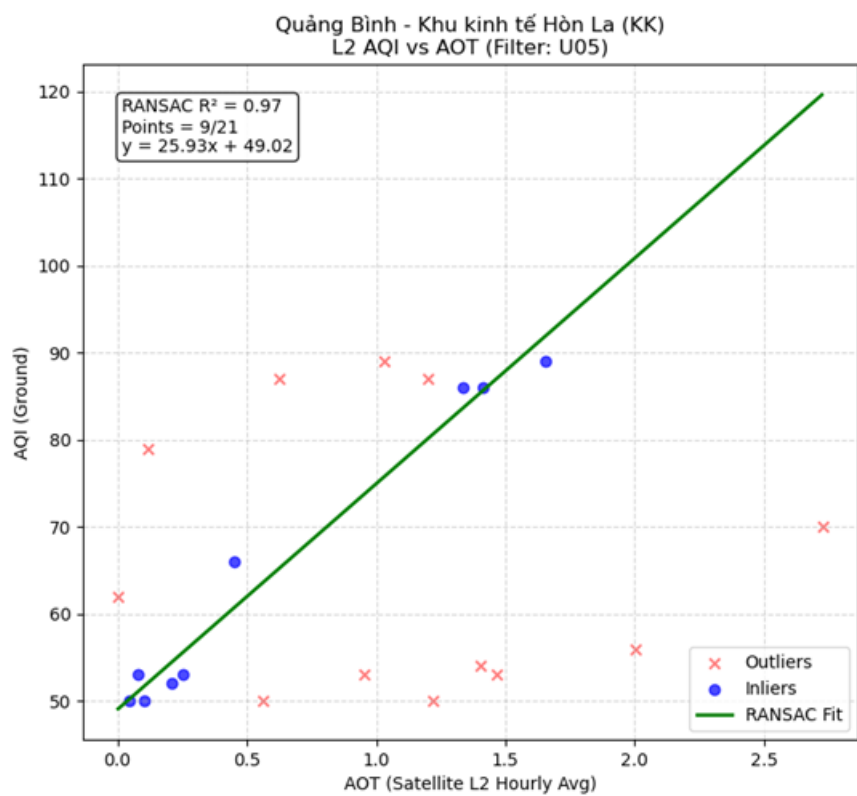


Figure 32: L2 RANSAC regression fit for Station 5.

HCM - Đ. Lê Hữu Kiều - P. Bình Trưng Tây - Quận 2 (Ngã ba Lê Hữu Kiều và Trương Văn Bang) (Kí
L2 AQI vs AOT (Filter: ALL)

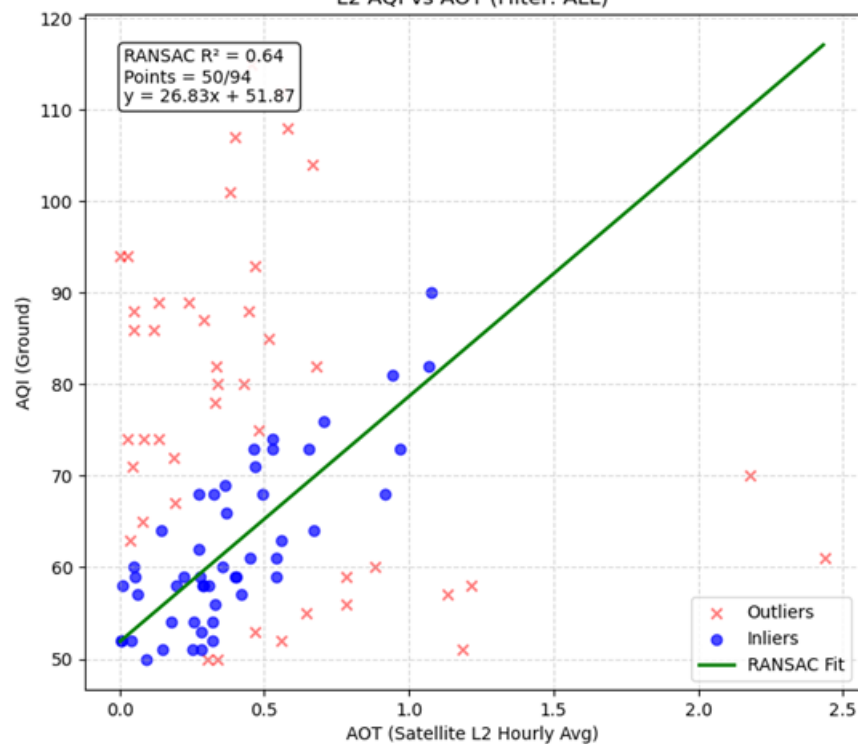


Figure 33: L2 RANSAC regression fit for Station 6.

HCM - Khu Liên cơ quan Bộ Tài Nguyên và Môi Trường - số 20 Đ. Lý Chính Thắng (KK)
L2 AQI vs AOT (Filter: U15)

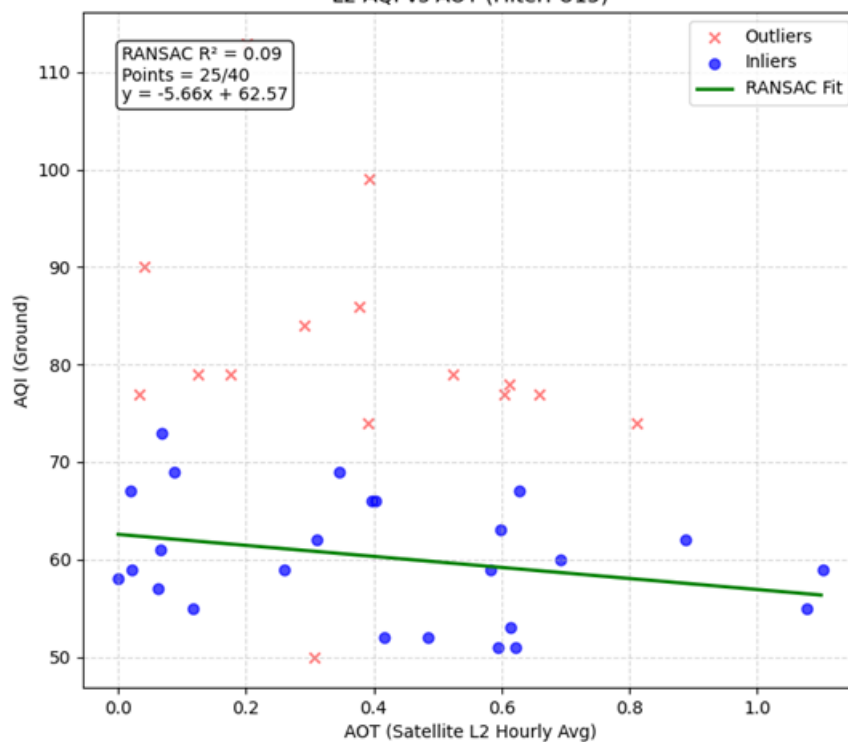


Figure 34: L2 RANSAC regression fit for Station 7.

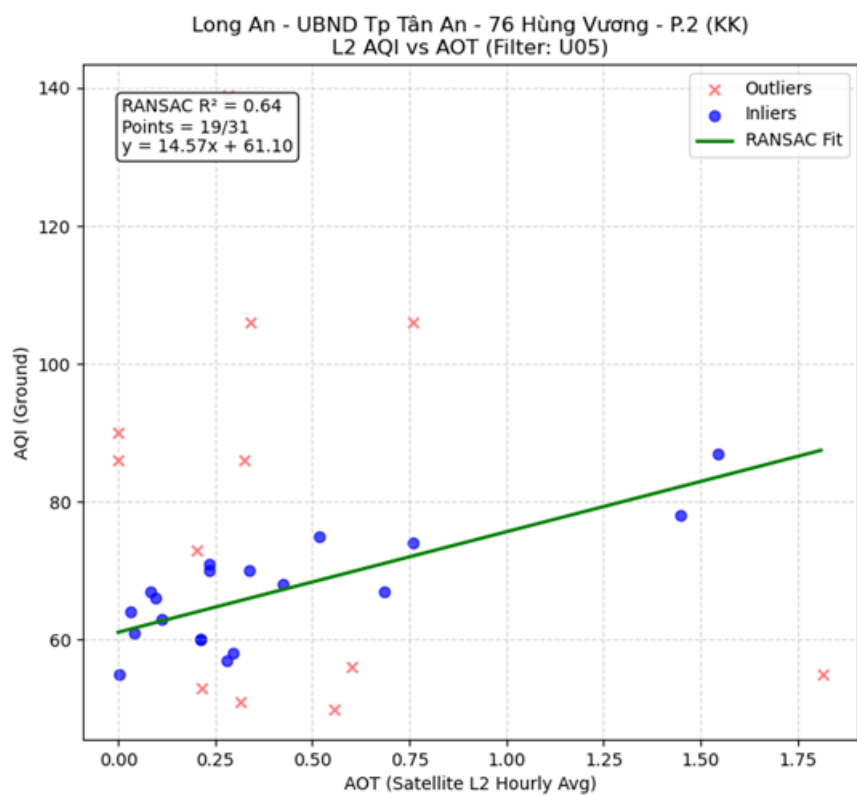


Figure 35: L2 RANSAC regression fit for Station 8.

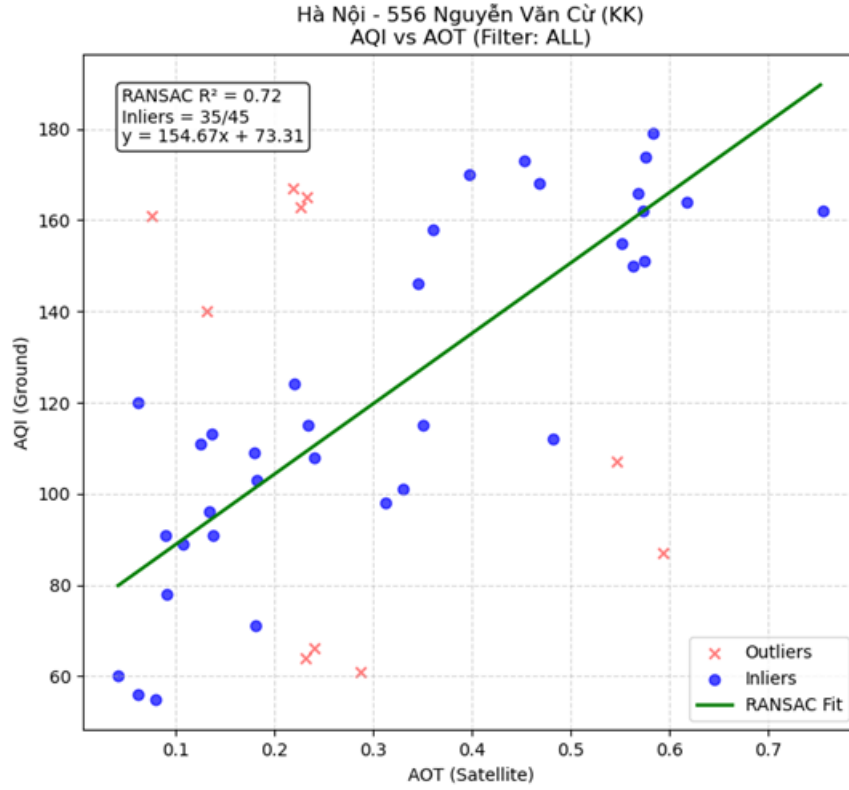


Figure 36: L2 RANSAC regression fit for Station 9.

B.2 L3 RANSAC Analysis

The Level 3 dataset demonstrated significantly improved data quality, with inlier retention rates consistently above 58%. The temporal aggregation inherent in L3 processing effectively removes transient atmospheric artifacts, resulting in more stable regression fits.

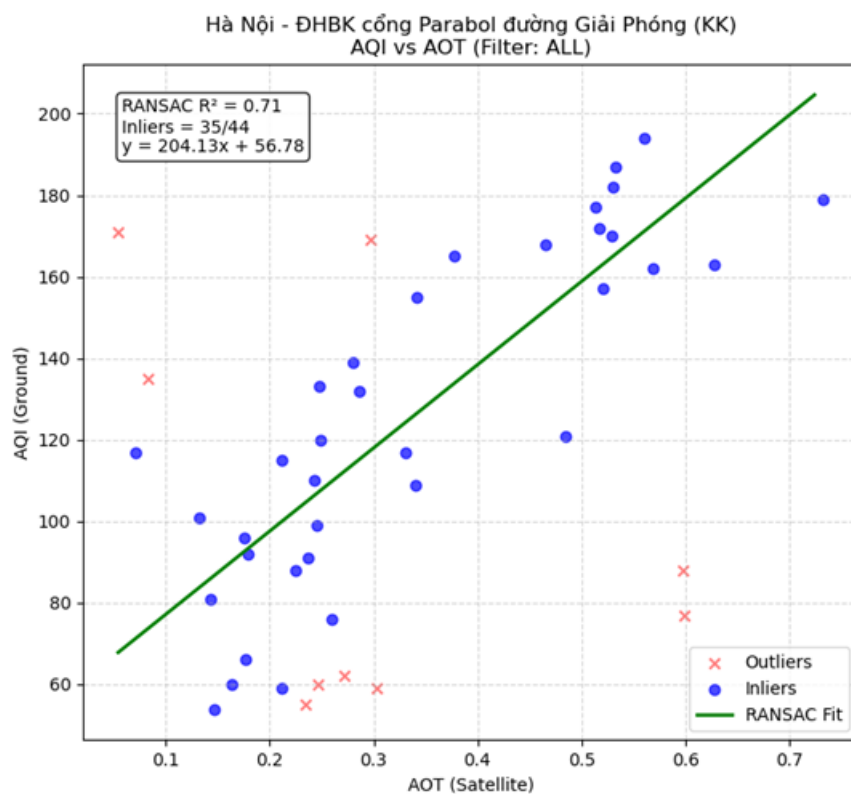


Figure 37: L3 RANSAC regression fit for Station 1.

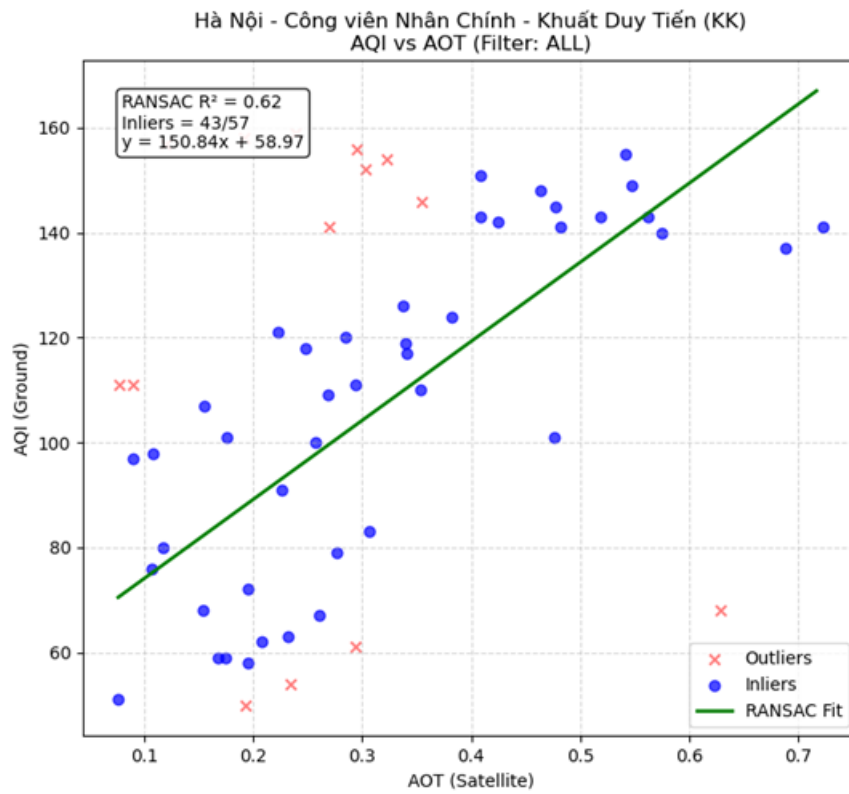


Figure 38: L3 RANSAC regression fit for Station 2.

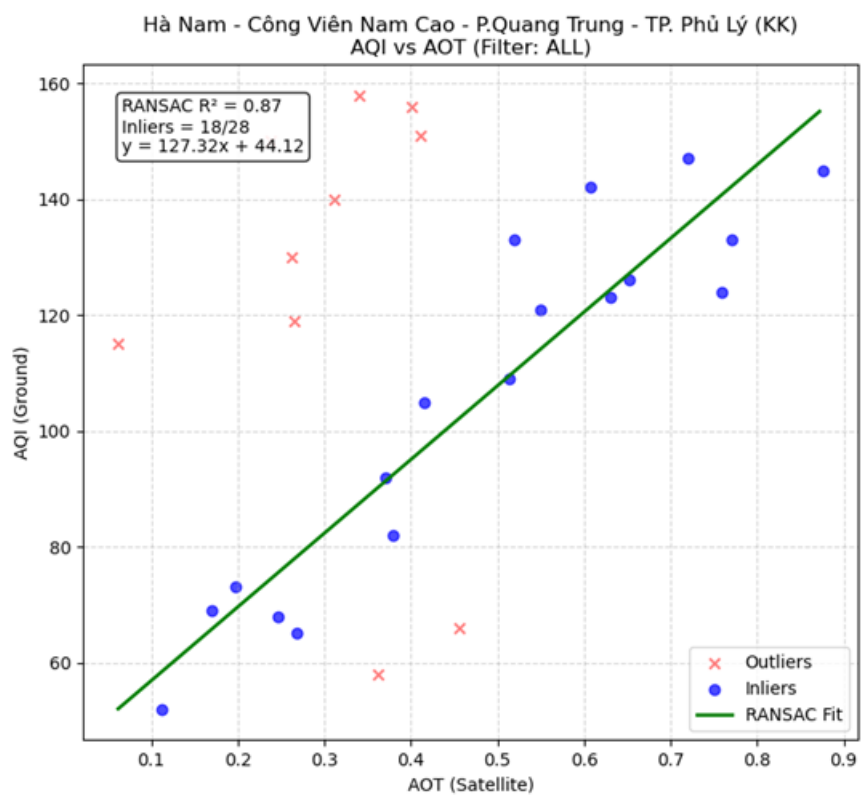


Figure 39: L3 RANSAC regression fit for Station 3.

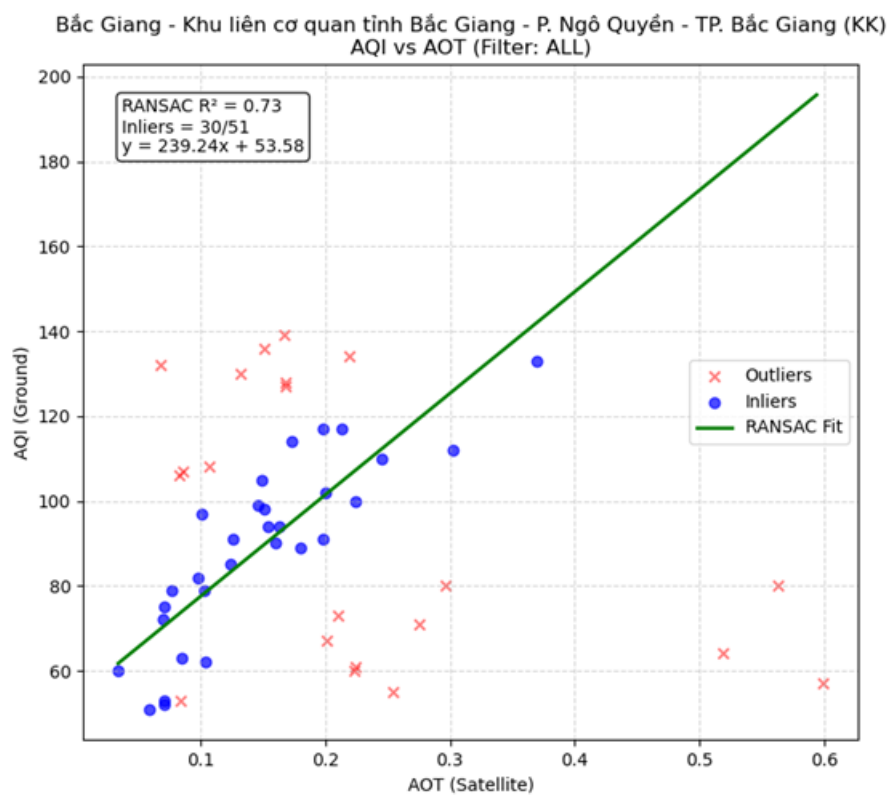


Figure 40: L3 RANSAC regression fit for Station 4.